Reviewer: Matthew P. Koenig
Date: 2/27/2026

PROJECT	DESIGNATION
1500342	1500342
CONTRACT	BRIDGE FILE
B-38321	044-70-06634 A

INDIANA DEPARTMENT OF TRANSPORTATION

STRUCTURE	TYPE	SPAN AND SKEW	OVER	STATION
044-70-06634 A	CONTINUOUS COMPOSITE STEEL BEAM BRIDGE	3 SPANS: 49'-9", 50'-0", 49'-9" SKEW: 15°00' RT.	MUD CREEK	617+95.00

BRIDGE REHABILITATION PLANS

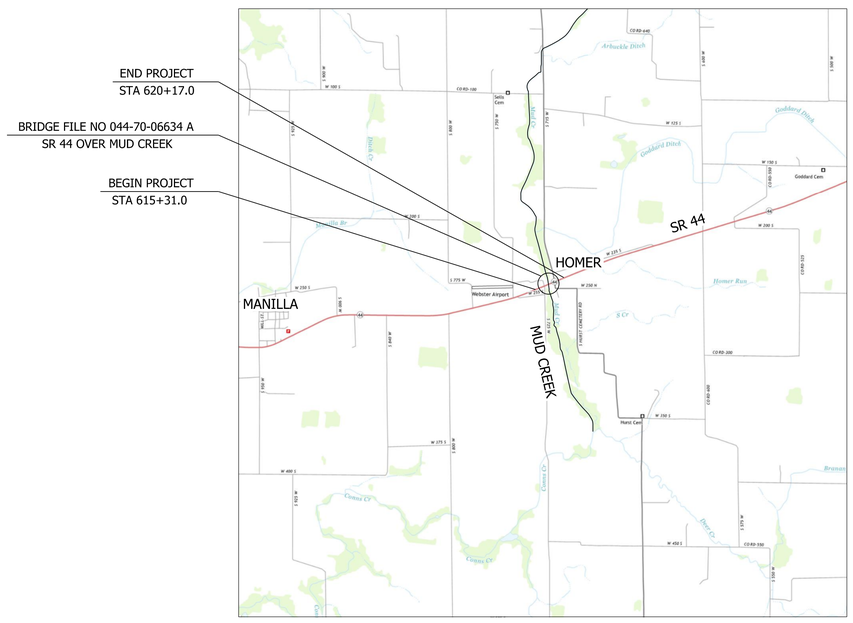
FOR SPANS OVER 20 FEET

ROUTE: SR 44 AT: RP 47 +64

PROJECT NO. 1500342 (P.E. R/W, CONST.)

NO ADDITIONAL RIGHT-OF-WAY
REQUIRED FOR THIS PROJECT

BRIDGE REHABILITATION ON SR 44 OVER MUD CREEK, LOCATED APPROXIMATELY 7.46 MILES
WEST OF US 52, IN SECTION 13, T-13-N, R-8-E, WALKER TOWNSHIP,
RUSH COUNTY, INDIANA.



LOCATION MAP
(RUSH COUNTY)

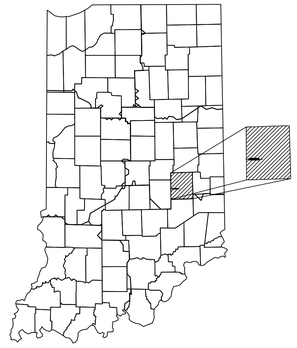
SCALE: 1" = 3000'

TRAFFIC DATA

A.A.D.T.	(2018)	4810	V.P.D.
A.A.D.T.	(2038)	5660	V.P.D.
D.H.V	(2038)	536	V.P.H.
DIRECTIONAL DISTRIBUTION		50.46	%
TRUCKS		25.87	% A.A.D.T.
		21.76	% D.H.V.

DESIGN DATA

DESIGN SPEED	50	M.P.H.
PROJECT DESIGN CRITERIA	3R (NON-FREEWAY)	
FUNCTIONAL CLASSIFICATION	MINOR ARTERIAL	
RURAL/URBAN	RURAL	
TERRAIN	LEVEL	
ACCESS CONTROL	NONE	



PROJECT LOCATIONS SHOWN BY
RUSH COUNTY

LATITUDE: 39° 34' 35" LONGITUDE: -85° 34' 44"

BRIDGE LENGTH: 0.028 MI.
ROADWAY LENGTH: 0.064 MI.
TOTAL LENGTH: 0.092 MI.
MAX. GRADE: 0.820 %

HUC: 05120205040070

INDIANA DEPARTMENT OF TRANSPORTATION
STANDARD SPECIFICATIONS DATED 2018
TO BE USED WITH THESE PLANS.

HNTB Corporation
The HNTB Companies
Infrastructure Solutions

111 Monument Circle
Suite 1200
Indianapolis, IN 46204

SKYLER R.P. COONEY — REGISTERED — No. 12006540 — STATE OF INDIANA — PROFESSIONAL ENGINEER

PLANS PREPARED BY: HNTB Indiana, Inc. (317) 636-4682 PHONE NUMBER

CERTIFIED BY: 08/22/2017 DATE

APPROVED FOR LETTING: INDIANA DEPARTMENT OF TRANSPORTATION DATE

BRIDGE FILE
044-70-06634 A
DESIGNATION
1500342

SURVEY BOOK	SHEETS
ELECTRONIC	1 of 34
CONTRACT	PROJECT
B-38321	1500342

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file: \\indx\00\269projects\64259 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-tit01.dgn

UTILITIES

COMMUNICATION:

FRONTIER COMMUNICATIONS OF INDIANA
DAN WHITE
221 W. WASHINGTON STREET
FAIRMOUNT, IN 46928
765-436-2206
DANIEL.WHITE@FRONTIERCORP.COM

NINE STAR CONNECT
TERRY MCDONNELL
2331 EAST 600 NORTH
MAXWELL, IN 46140
317-323-2082
TMCDONNELL@TEAMHANCOCK.COM

SEWER AND WATER:

WESTERN RUSH COUNTY REGIONAL
SEWER AND WATER DISTRICT
TERRY ROGERS
6682 WEST 235 SOUTH
HOMER, IN 46146
765-663-5871
TROGERS@DECATURWB.COM

Indiana 811
Know what's below.
Call before you dig.

INDEX

SHEET NO.	SUBJECT
1	TITLE
2	INDEX
3-5	DETOUR DETAILS
6	APPROACH ROADWAY DETAILS
7-8	EROSION CONTROL
9	LAYOUT
10-11	GENERAL PLAN
12-13	REMOVAL DETAILS
14-15	END BENT DETAILS
16	PIER DETAILS
17	FRAMING PLAN
18-21	STRUCTURAL STEEL DETAILS
22	BEARING DETAILS
23-28	SUPERSTRUCTURE DETAILS
29-30	SCREEDS
31	RAILING DETAILS
32-33	APPROACH SLAB DETAILS
34	BRIDGE SUMMARY

REVISIONS

SHEET NO.	DATE	REVISED

SKYLER R.P. COMBS
REGISTERED
No. 11200658
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ 08/22/2017
DESIGN ENGINEER DATE

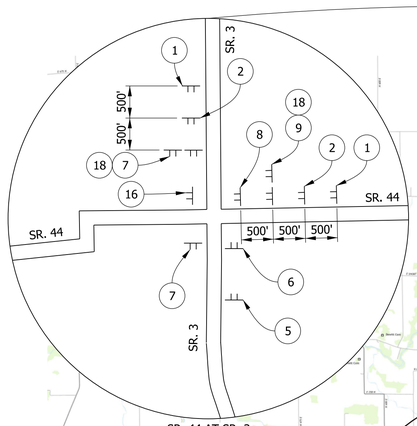
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DRAWN: CDC 07/2016
CHECKED: MEW 08/2016
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INDIANA
DEPARTMENT OF TRANSPORTATION

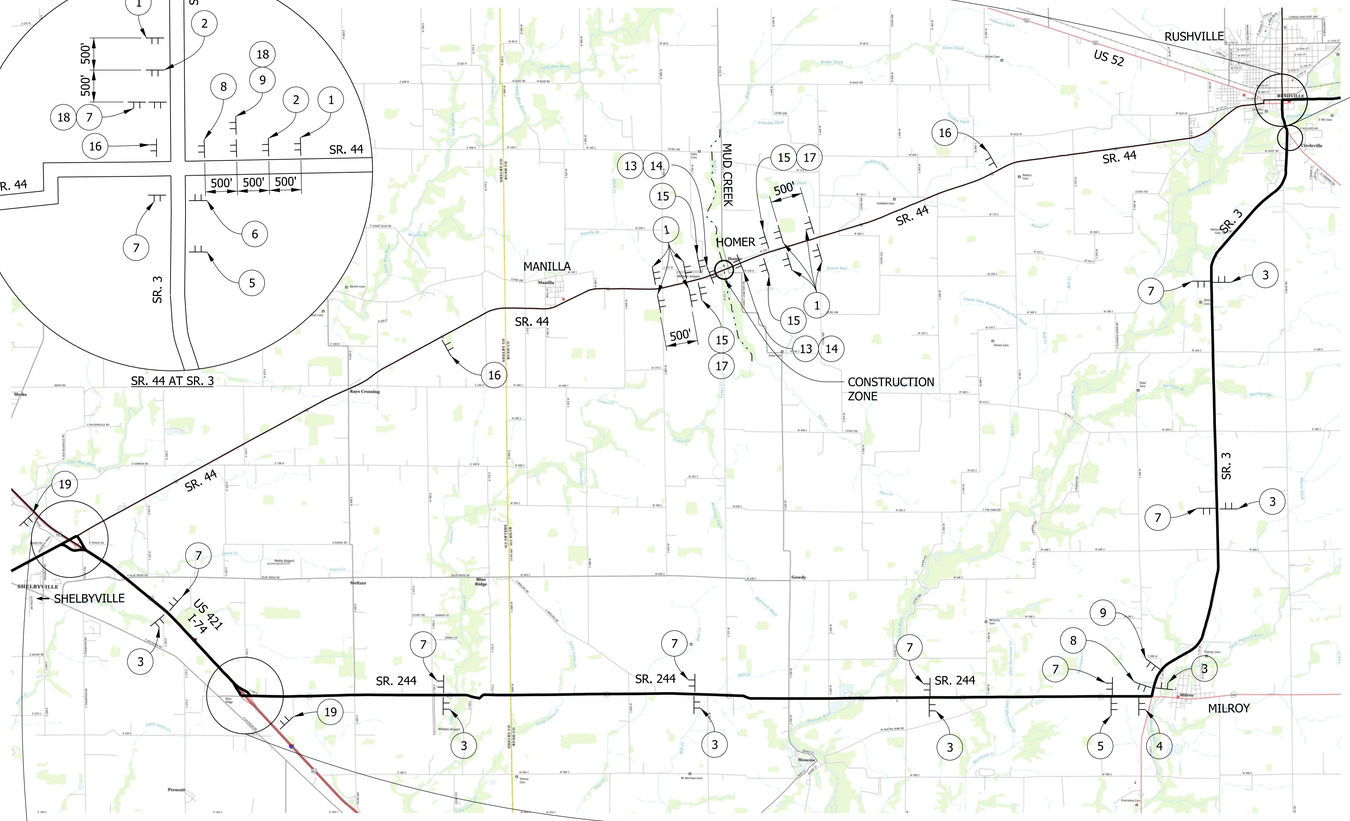
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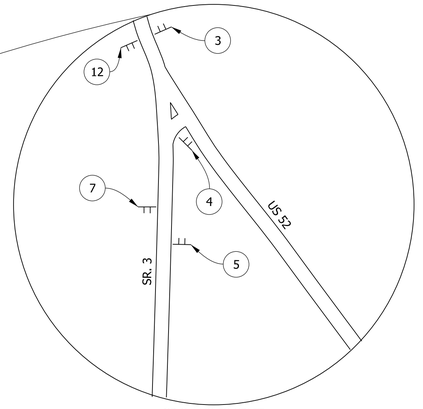
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VERTICAL SCALE	DESIGNATION
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SURVEY BOOK	SHEETS
ELECTRONIC	2 of 34
CONTRACT	PROJECT
B-38321	1500342

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SR. 44 AT SR. 3





SR 3 / US 52 SPLIT

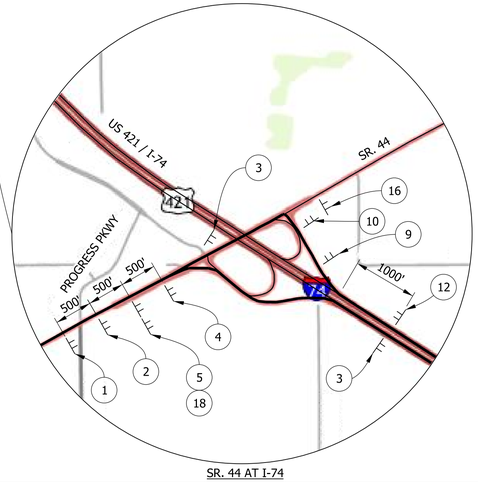
MOT SUMMARY		
ITEM	UNITS	TOTALS
DETOUR ROUTE MARKER ASSEMBLIES	EA	43
CONSTRUCTION SIGN, A	EA	20
CONSTRUCTION SIGN, C	EA	3
TEMPORARY PANEL SIGNS	SFT	110
TEMPORARY PANEL SIGN SUPPORTS	LFT	80
ROAD CLOSURE SIGN ASSEMBLY	EA	4
BARRICADE, III-A	LFT	132
BARRICADE, III-B	LFT	48
MAINTAINING TRAFFIC	LS	1

** INCLUDES 2 XG20-5 ROUTE CLOSURE NOTICE SIGNS TO BE LOCATED AS DIRECTED BY ENGINEER.

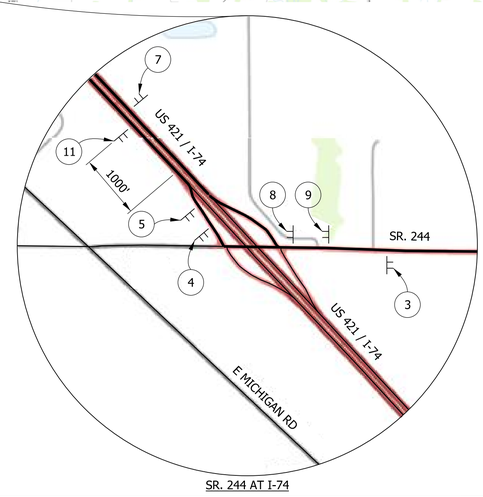
ROAD CLOSED AHEAD — XW20-3 — 1

DETOUR AHEAD — XW20-2 ** — 2

DETOUR EAST INDIANA 44	XM4-8 M3-2 M1-5 M6-3	3
DETOUR EAST INDIANA 44	XM4-8 M3-2 M1-5 M6-1 (R OR L)	4
DETOUR EAST INDIANA 44	XM4-8 M3-2 M1-5 M5-1 (R OR L)	5
END DETOUR EAST INDIANA 44	M4-8b XM4-8 M3-2 M1-5 M6-1 R	6
DETOUR WEST INDIANA 44	XM4-8 M3-4 M1-5 M6-3	7
DETOUR WEST INDIANA 44	XM4-8 M3-4 M1-5 M6-1 (R OR L)	8
DETOUR WEST INDIANA 44	XM4-8 M3-4 M1-5 M5-1 (R OR L)	9
END DETOUR WEST INDIANA 44	M4-8b XM4-8 M3-4 M1-5 M6-1 L	10
DETOUR EAST INDIANA 44	XM4-8 M3-2 M1-5 M6-2S	11
DETOUR WEST INDIANA 44	XM4-8 M3-4 M1-5 M6-2S	12



SR. 44 AT I-74



SR. 244 AT I-74

TYPE III-A BARRICADE — 13

ROAD CLOSED — R11-2 — 14

TYPE III-B BARRICADE — 15

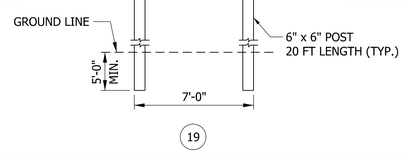
BRIDGE OUT XX MILES AHEAD LOCAL TRAFFIC ONLY — R11-3b — 16

ROAD CLOSED TO THRU TRAFFIC — R11-4 — 17

SR 44 CLOSED IN HOMER — 18

SR 44 CLOSED AT HOMER DETOUR EXIT 119

SR 44 CLOSED (DATE) — XG20-5 **



19

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SKYLER R.P. COMBS — REGISTERED PROFESSIONAL ENGINEER — No. 11200658 — STATE OF INDIANA

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: CDC 07/2016 — DRAWN: CDC 07/2016

CHECKED: MEW 07/2016 — CHECKED: MEW 07/2016

INDIANA
DEPARTMENT OF TRANSPORTATION

DETOUR DETAILS

SCALE	BRIDGE FILE
NTS	044-70-06634 A
VERTICAL SCALE	DESIGNATION
NTS	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	3 of 34
CONTRACT	PROJECT
B-38321	1500342



SIGN NUMBER	18
WIDTH x HGHT.	6'-6" x 2'-6"
BORDER WIDTH	1"
CORNER RADIUS	2"
MOUNTING	Ground
BACKGROUND	TYPE: Reflective
	COLOR: Orange
LEGEND/BORDER	TYPE: Reflective
	COLOR: Black/Black

SYMBOL	ROT	X	Y	WID	HT

Dimensions are in inches.tenths

Letter locations are panel edge to lower left corner

LETTER POSITIONS (X)																																		LENGTH	SERIES/SIZE
S	R		4	4		C	L	O	S	E	D																									D 2000
9	14.1	18.2	24.2	29.5	34	40	45.4	49.9	55.1	60.2	64.9																								60	6
I	N		H	O	M	E	R																													D 2000
19.8	22.2	26.3	32.3	37.7	43.3	49.4	54.1																												38.4	6

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SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER 08/22/2017 DATE

DESIGNED: JGO 06/2017 DRAWN: JGO 06/2017

CHECKED: HER 06/2017 CHECKED: HER 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

DETOUR DETAILS

SCALE	BRIDGE FILE
NTS	044-70-06634 A
VERTICAL SCALE	DESIGNATION
NTS	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	4 of 34
CONTRACT	PROJECT
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SR 44 CLOSED
AT HOMER
DETOUR EXIT 119

SIGN NUMBER	19
WIDTH x HGHT.	11'-0" x 5'-0"
BORDER WIDTH	2"
CORNER RADIUS	9"
MOUNTING	Ground
BACKGROUND	TYPE: Reflective
	COLOR: Orange
LEGEND/BORDER	TYPE: Reflective
	COLOR: Black/Black

SYMBOL	ROT	X	Y	WID	HT

Dimensions are in inches.tenths

Letter locations are panel edge to lower left corner

LETTER POSITIONS (X)																																		LENGTH	SERIES/SIZE
S	R		4	4		C	L	O	S	E	D																									EM 2000
19.4	27.8	34.3	42.3	51.1	58.6	66.6	74.8	81.7	90.1	98.6	106.2																								93.3	8
A	T		H	O	M	E	R																													EM 2000
34.3	43	48.9	56.9	65.3	74	83.6	91.2																												63.4	8
D	E	T	O	U	R		E	X	I	T		1	1	9																					EM 2000
15	23.4	30.2	37.2	45.9	54.6	61.1	69.1	76.1	84.8	87.8	93.7	101.7	106.2	110.5																					102	8

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SKYLER R.P. COMBS REGISTERED No. 10120065 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ 08/22/2017
DESIGN ENGINEER DATE

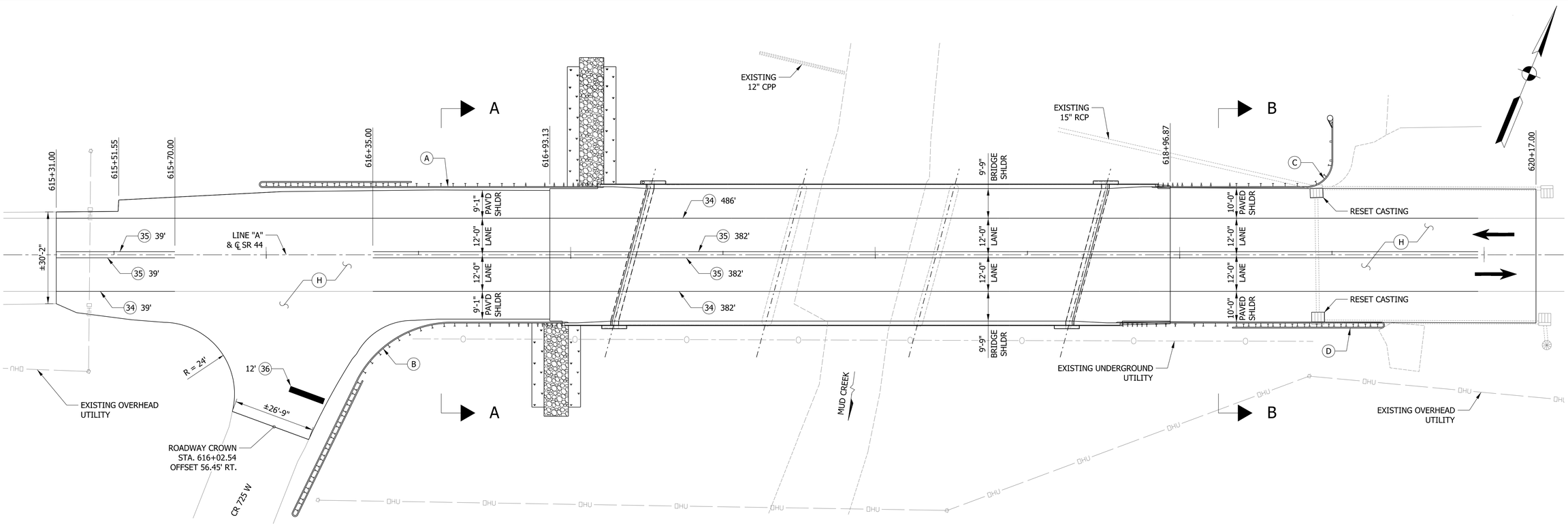
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CHECKED: HER 06/2017 CHECKED: HER 06/2017

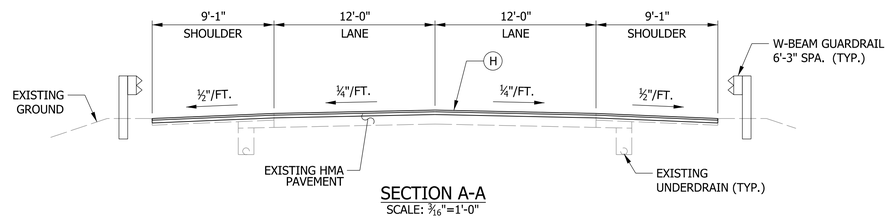
INDIANA
DEPARTMENT OF TRANSPORTATION

DETOUR DETAILS

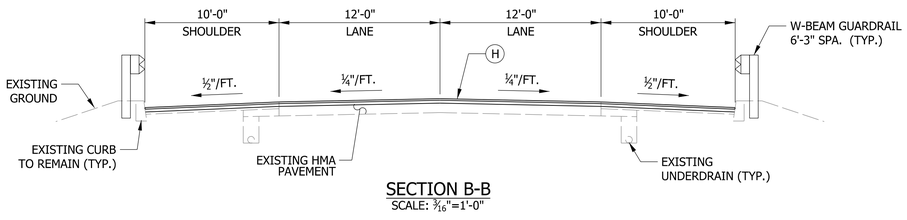
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VERTICAL SCALE	DESIGNATION
NTS	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	5 of 34
CONTRACT	PROJECT
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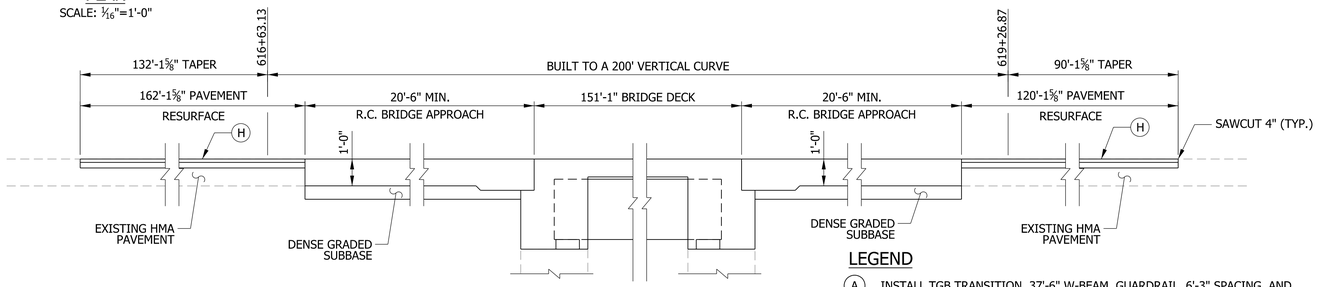
PLAN
SCALE: 1/16"=1'-0"



SECTION A-A
SCALE: 3/16"=1'-0"



SECTION B-B
SCALE: 3/16"=1'-0"



LONGITUDINAL SECTION
SCALE: 3/8"=1'-0"

NOTE: MAINTAIN TRAFFIC ON S CR 725 W THROUGHOUT CONSTRUCTION.

LEGEND

- (A) INSTALL TGB TRANSITION, 37'-6" W-BEAM, GUARDRAIL, 6'-3" SPACING, AND OS END TREATMENT.
- (B) INSTALL TGB TRANSITION, 12'-6" W-BEAM, GUARDRAIL, 6'-3" SPACING, MODIFIED CONNECTOR SYSTEM TYPE 2 (R = 35' WITH 7 POSTS), AND TYPE OS END TREATMENT.
- (C) INSTALL TGB TRANSITION, 25'-0" W-BEAM, GUARDRAIL, 6'-3" SPACING, TERMINAL SYSTEM TYPE 2 (R = 8.5'), AND TYPE 5 ANCHOR.
- (D) INSTALL TGB TRANSITION, 12'-6" W-BEAM, GUARDRAIL, 6'-3" SPACING, AND OS END TREATMENT.
- (H) TRANSITION MILLING AND 165 LB/SYS QC/QA-HMA, 3, 70, SURFACE, 9.5 MM ON 275 LB/SYS QC/QA-HMA, 3, 70, INTERMEDIATE, 19.0 MM.
- (34) LINE, PAINT, SOLID, WHITE, 4 IN.
- (35) LINE, PAINT, SOLID, YELLOW, 4 IN.
- (36) TRANSVERSE MARKING, PAINT, STOP LINE, WHITE, 24 IN.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MEW 08/2016 DRAWN: MEW 08/2016

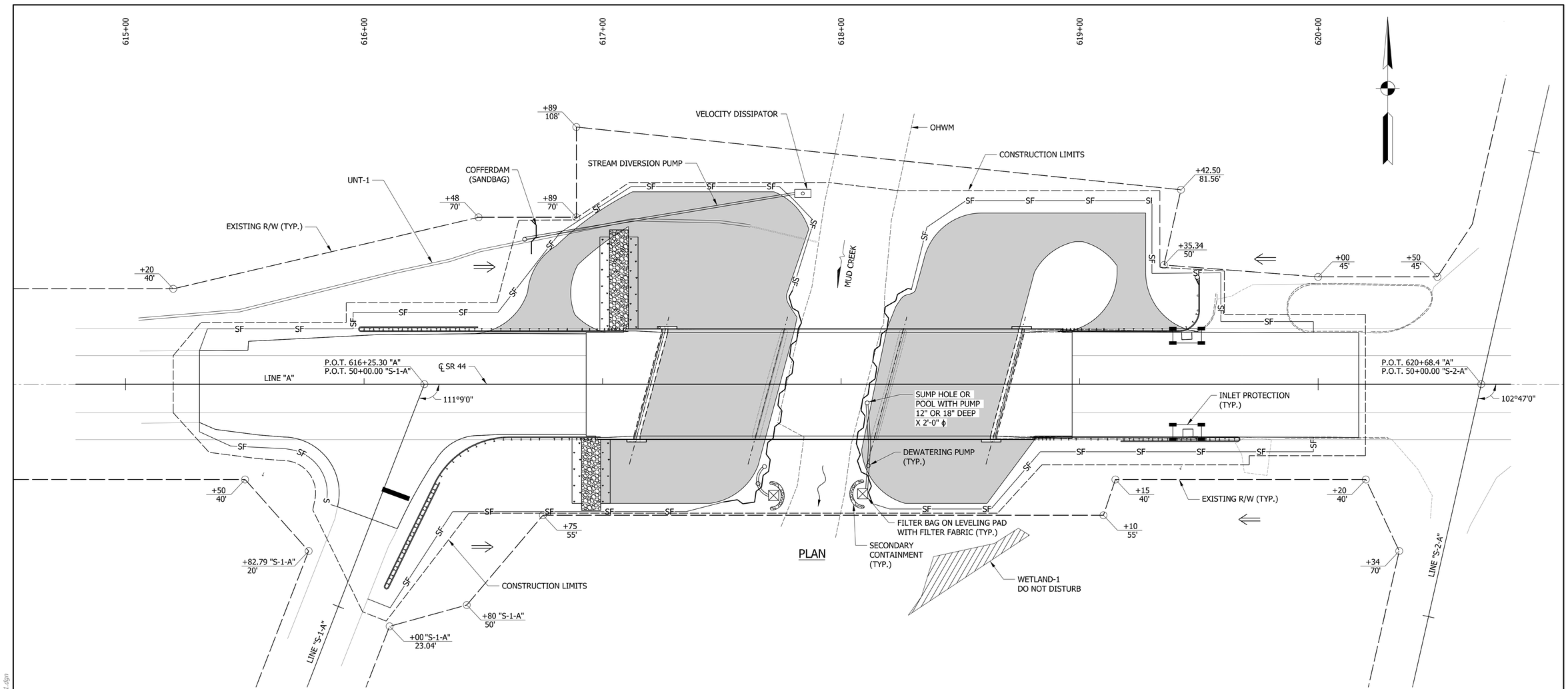
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INDIANA
DEPARTMENT OF TRANSPORTATION

APPROACH ROADWAY DETAILS

SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	6 of 34
CONTRACT	PROJECT
B-38321	1500342

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PLAN

EROSION CONTROL SILT FENCE *	
LOCATION	LENGTH (LFT)
615+31 TO 617+92 LT LINE "A"	360
615+31 TO 615+89 RT LINE "A"	80
616+14 TO 617+77 RT LINE "A"	220
618+05 TO 619+98 RT LINE "A"	240
618+18 TO 619+98 LT LINE "A"	285
TOTAL	1185

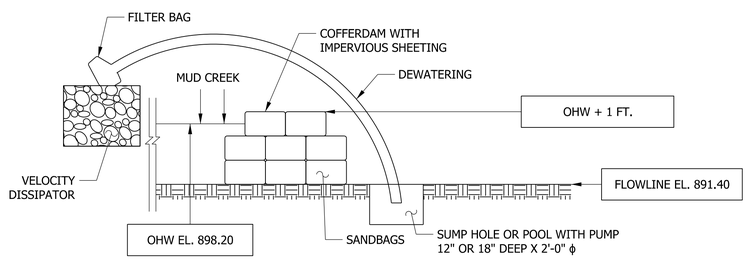
* FOR INFORMATION ONLY. ESTIMATED QUANTITY
INCLUDED IN STORM WATER MANAGEMENT BUDGET.

SEDIMENT REMOVE *	
BRIDGE	CYS
EAST CORNERS	1
WEST CORNERS	1

CONSTRUCTION * ENTRANCES	
NO. 2 STONE	TEMP. GEOTEXTILES
TONS	SYS
300	705

MULCHED SEEDING, R	
BRIDGE	SYS
EAST CORNERS	250
WEST CORNERS	250

TEMPORARY INLET PROTECTION *	
LOCATION	EACH
619+45 LINE "A"	1
619+45 LINE "A"	1
TOTAL	2



DEWATERING / WORK AREA
NOT TO SCALE

LEGEND

OHW	ORDINARY HIGH WATER
-SF-	TEMPORARY SILT FENCE
	TEMPORARY CONSTRUCTION ACCESS
	WETLAND
	OFFSITE DRAINAGE POINT OF ENTRY
	STORM WATER EXIT (OPEN CHANNEL)
	TEMPORARY COFFERDAM

SKYLER R.P. COMBS
REGISTERED
No. 11200658
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MRS 02/17 DRAWN: MRS 02/17
CHECKED: SRPC 02/17 CHECKED: SRPC 06/17

INDIANA
DEPARTMENT OF TRANSPORTATION

EROSION CONTROL PLAN

SCALE	BRIDGE FILE
1' = 20'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
1' = 20'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	7 of 34
CONTRACT	PROJECT
B-38321	1500342

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SOIL CLASSIFICATIONS	
MAP SYMBOL	MAP UNIT NAME
CrA	CROSBY SILT LOAM, 0 TO 2 PERCENT SLOPES
MpB2	MIAMIAN SILT LOAM, NEW CASTLE TILL PLAIN, 2 TO 6 PERCENT SLOPES, ERODED
Sh	SHOALS SILT LOAM, 0 TO 2 PERCENT SLOPES, FREQUENTLY FLOODED, VERY BRIEF DURATION
Tr	TREATY SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES

SKYLER R.P. COOMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER 08/22/2017 DATE

DESIGNED: MRS 02/17 DRAWN: MRS 02/17

CHECKED: SRPC 06/17 CHECKED: SRPC 06/17

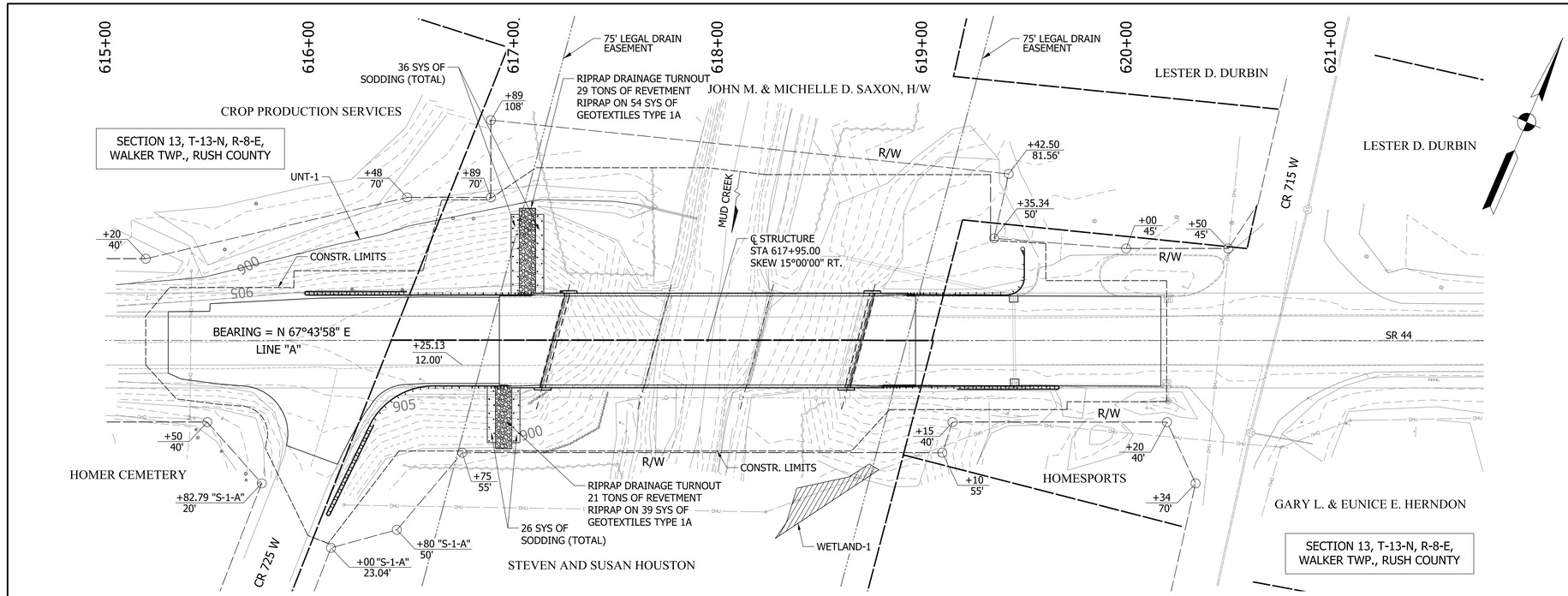
INDIANA
DEPARTMENT OF TRANSPORTATION

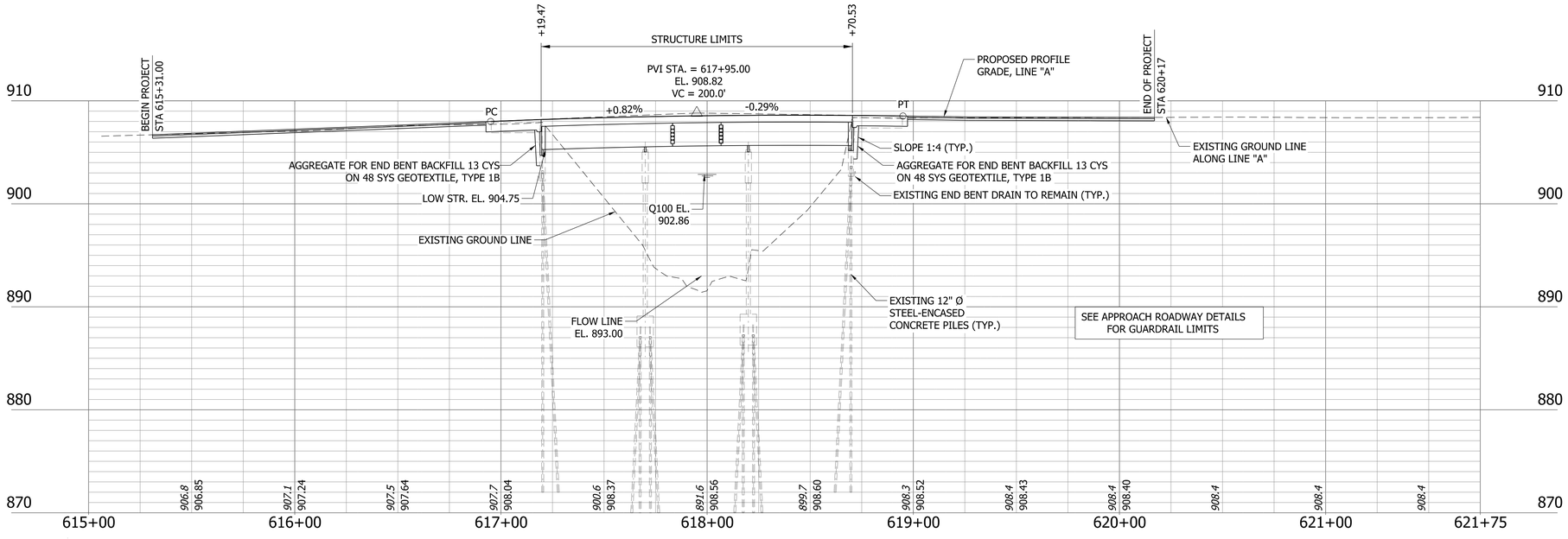
EROSION CONTROL
SOILS MAP

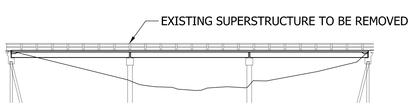
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VERTICAL SCALE	DESIGNATION
1" = 60'-0"	1500342

SURVEY BOOK	SHEETS
ELECTRONIC	8 of 34
CONTRACT	PROJECT
B-38321	1500342

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EXISTING STRUCTURE

THE EXISTING STRUCTURE IS A 3 SPAN CONTINUOUS COMPOSITE PRESTRESSED CONCRETE I-BEAM BRIDGE BUILT IN 1983.
PORTIONS OF EXISTING STRUCTURE TO REMAIN IN PLACE.

HYDRAULIC SCOUR DATA

Q100 DISCHARGE	3540 CFS
Q100 ELEV.	902.86 FT
VELOCITY AT Q100	4.39 FT/SEC
SCOUR DEPTH (CONTRACTION)	0.61 FT
SCOUR DEPTH (TOTAL)	6.38 FT
LOW SCOUR ELEV.	886.62 FT

EARTHWORK TABULATION

COMMON EXCAVATION	70 CYS
FOUNDATION EXCAVATION UNCLASSIFIED	40 CYS

NOTES

1. EXISTING BRIDGE ELEVATIONS SHALL BE CHECKED BY THE CONTRACTOR TO ENSURE PROPER FIT OF NEW CONCRETE TO EXISTING CONCRETE. EXISTING ELEVATIONS SHOWN IN THESE PLANS ARE BASED ON ELEVATIONS SHOWN IN PLAN SET 44-70-6634, DATED MAY 16, 1980 AND LOWERED BY 0.57' TO MORE CLOSELY MATCH THE CURRENT SURVEY.
2. ALL CONTOURS ARE EXISTING.
3. RIGHT OF WAY BASED ON LINE "A"
4. ALL RIGHT OF WAY TO BE AS SHOWN.

CONTINUOUS COMPOSITE
STEEL BEAM BRIDGE
3 SPANS: 49'-9", 50'-0", 49'-9"
SKEW: 15°00'00" RT.
43'-6" CLEAR ROADWAY
SR 44 OVER MUD CREEK
RUSH COUNTY

SKYLER R.P. COOMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

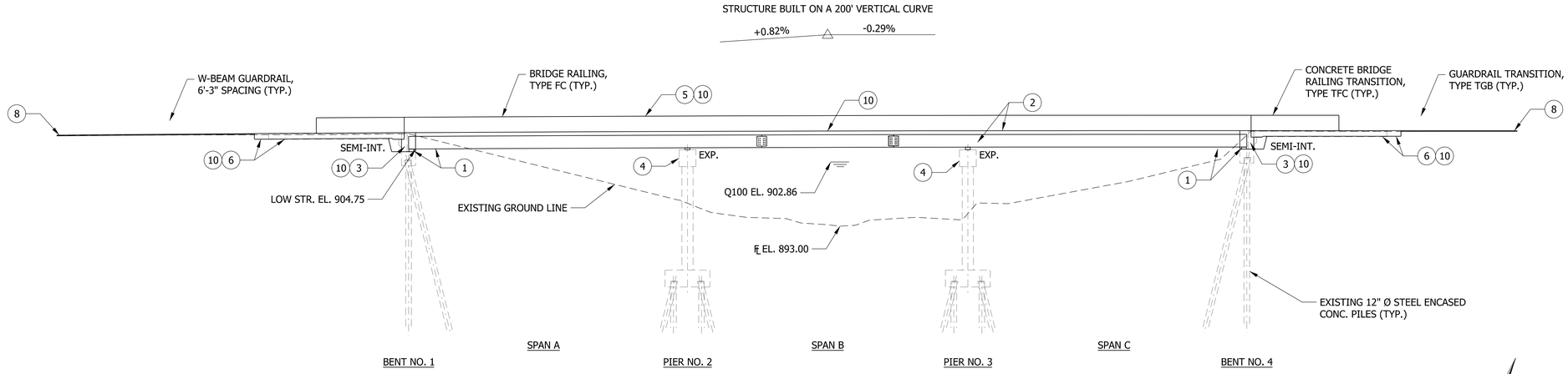
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CHECKED: SRPC 08/2016 CHECKED: SRPC 08/2016

INDIANA
DEPARTMENT OF TRANSPORTATION

LAYOUT

SCALE	BRIDGE FILE
1" = 30'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
1" = 5'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	9 of 34
CONTRACT	PROJECT
B-38321	1500342

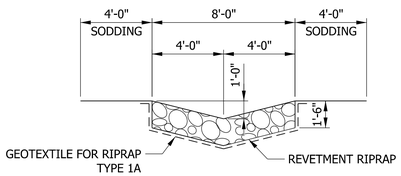
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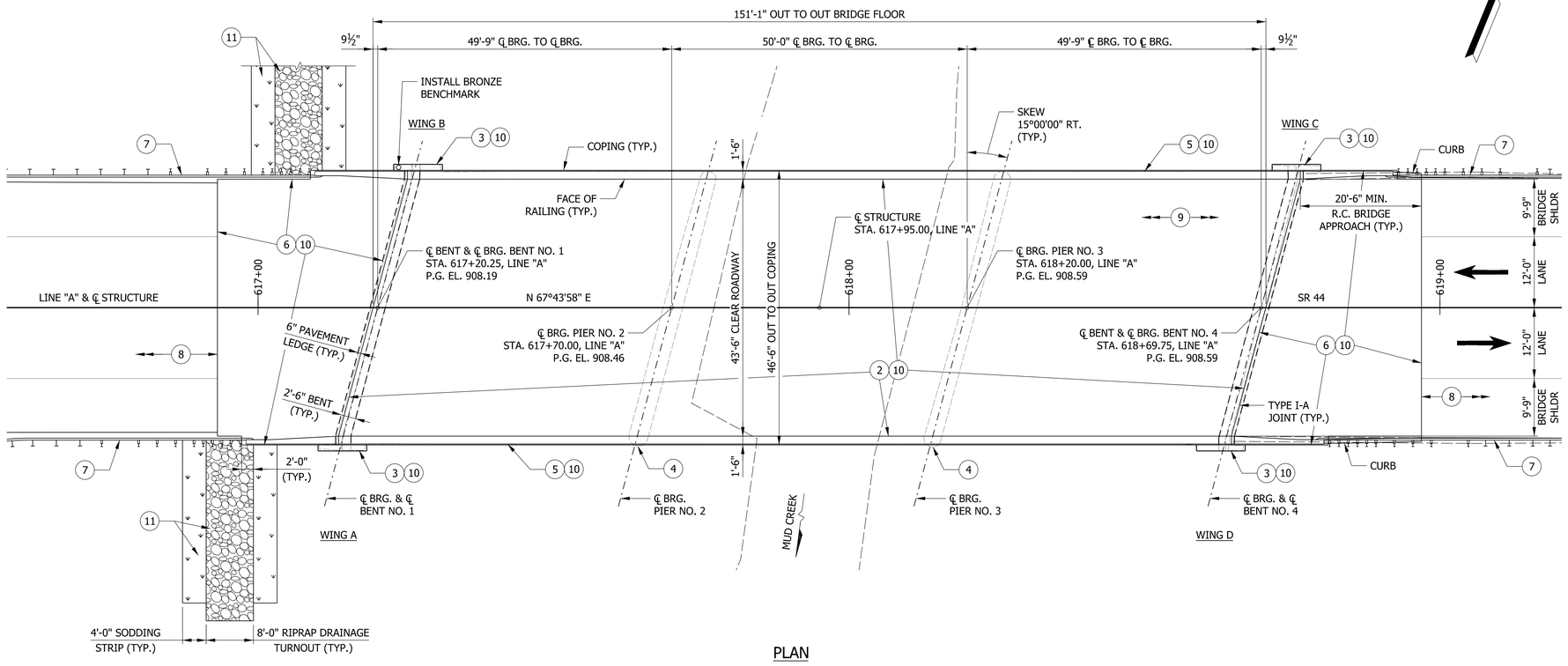
ELEVATION

LEGEND

1. REMOVE EXISTING SUPERSTRUCTURE INCLUDING EXISTING BEARINGS.
2. CONSTRUCT NEW CONTINUOUS COMPOSITE STEEL BEAM SUPERSTRUCTURE.
3. REMOVE PORTIONS OF ABUTMENTS AND WINGWALLS DOWN TO THE BRIDGE SEAT. RECONSTRUCT AS SEMI-INTEGRAL END BENTS.
4. REPAIR DELAMINATED AREAS OF THE PIERS BY REMOVING UNSOUND CONCRETE AND PATCHING.
5. CONSTRUCT NEW TYPE FC CONCRETE BRIDGE RAILING AND INSTALL BARRIER DELINEATORS ON TOP OF RAIL AT 40'-0" MAX. SPACING.
6. REMOVE AND REPLACE EXISTING APPROACH SLABS WITH NEW FULL WIDTH APPROACH SLABS.
7. REMOVE AND REPLACE SUBSTANDARD APPROACH GUARDRAIL, GUARDRAIL TRANSITIONS, AND END TREATMENTS.
8. MILL AND RESURFACE THE APPROACH PAVEMENT.
9. RESTRIPE PAVEMENT MARKINGS WITHIN PROJECT LIMITS.
10. SURFACE SEAL ALL EXPOSED FACES OF THE BRIDGE DECK, COPINGS, END BENTS, BARRIER RAILING AND REINFORCED CONCRETE BRIDGE APPROACHES.
11. INSTALL RIPRAP TURNOUT AND SOD STRIPS.



RIPRAP DRAINAGE TURNOUT
TYPICAL SECTION
SCALE: ¼" = 1'-0"



PLAN

CONTINUOUS COMPOSITE
STEEL BEAM BRIDGE
3 SPANS: 49'-9", 50'-0", 49'-9"
SKEW: 15°00'00" RT.
43'-6" CLEAR ROADWAY
SR 44 OVER MUD CREEK
RUSH COUNTY

SKYLER R.P. CONDES — REGISTERED — No. 11200658 — STATE OF INDIANA — PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

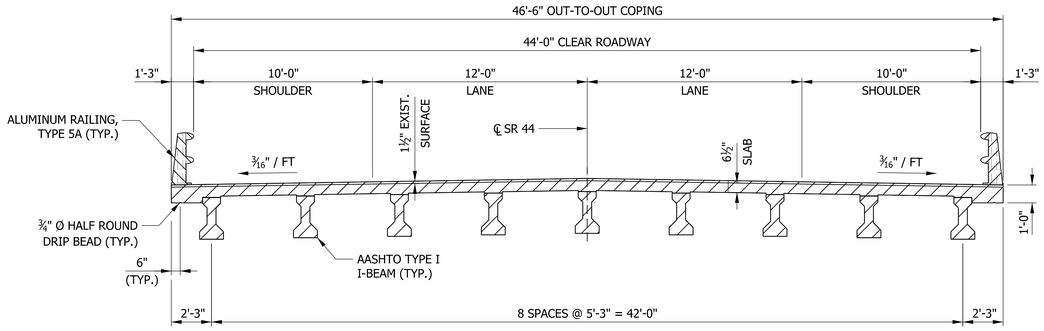
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CHECKED: SRPC 08/2016 CHECKED: SRPC 08/2016

INDIANA
DEPARTMENT OF TRANSPORTATION

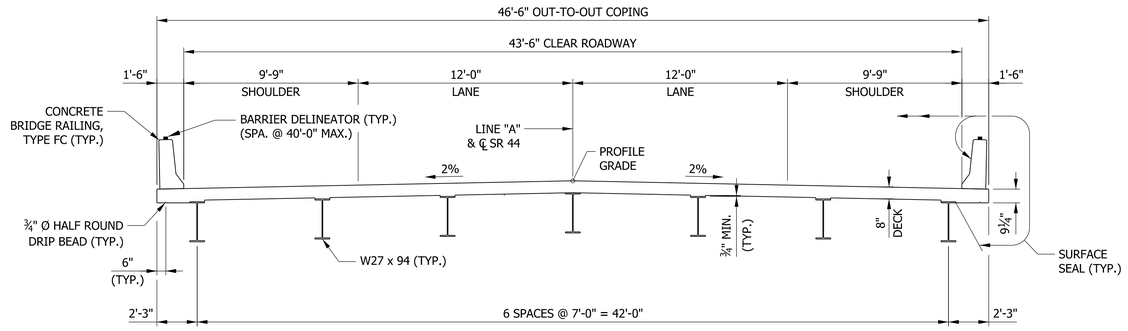
GENERAL PLAN
PROPOSED STRUCTURE

SCALE	BRIDGE FILE
3/32" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
3/32" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	10 of 34
CONTRACT	PROJECT
B-38321	1500342

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EXISTING TYPICAL SECTION
SCALE: ¼" = 1'-0"



PROPOSED TYPICAL SECTION
SCALE: ¼" = 1'-0"

GENERAL NOTES

PLANS FOR THE EXISTING BRIDGE ARE ON FILE IN THE RECORDS UNIT OF THE INDIANA DEPARTMENT OF TRANSPORTATION AS BRIDGE FILE 44-70-6634.

WHERE NEW WORK IS TO BE FITTED TO OLD WORK, THE CONTRACTOR SHALL CHECK ALL DIMENSIONS AND CONDITIONS IN THE FIELD AND REPORT ALL ERRORS OR DISCREPANCIES TO THE ENGINEER AND ASSUME RESPONSIBILITY FOR THEIR CORRECTNESS AND FIT OF THE NEW PART TO THE OLD.

REINFORCING BAR COVER SHALL BE 2½" MIN. IN TOP AND 1" MIN. IN BOTTOM AND 2" EVERYWHERE ELSE UNLESS OTHERWISE NOTED.

EPOXY RESIN ADHESIVE SHALL BE USED WHERE NEW CONCRETE ABUTS EXISTING CONCRETE.

CLEAN AND SURFACE SEAL ALL EXPOSED FACES OF THE ABUTMENTS / WINGWALLS, BARRIER RAILING, REINFORCED CONCRETE BRIDGE APPROACH, DECK, COPING, AND END BENTS.

ALL PAVEMENT MARKINGS DISTURBED DURING CONSTRUCTION SHALL BE REPLACED IN KIND.

DESIGN DATA

ORIGINAL STRUCTURE
DESIGNED FOR HS 20-44 LOADING IN ACCORDANCE WITH 1977 AASHTO SPECIFICATIONS AND 1978 INTERIMS. DEAD LOAD ACCOUNTED FOR ACTUAL PLUS 35 PSF FOR A FUTURE WEARING SURFACE.

LIVE LOAD
SUPERSTRUCTURE DESIGNED FOR HL-93 LOADING, IN ACCORDANCE WITH AASHTO LRFD BRIDGE DESIGN SPECIFICATIONS, 6TH EDITION, 2012 AND INTERMS THROUGH 2013.

DEAD LOAD
ACTUAL WEIGHT PLUS 35 PSF FOR FUTURE WEARING SURFACE AND 15 PSF FOR PERMANENT METAL DECK FORMS.

FLOOR SLAB
DESIGNED WITH A 7½" STRUCTURAL DEPTH PLUS ½" SACRIFICIAL WEARING SURFACE.

LEGEND

INDICATES LIMITS OF REMOVAL

DESIGN STRESSES

CONCRETE
CLASS "C" CONCRETE f'c=4,000 psi

REINFORCING STEEL
REINFORCING STEEL fy=60,000 psi

STRUCTURAL STEEL
ASTM A709 GRADE 50W fy=50,000 psi

CONSTRUCTION LOADING
THE EXTERIOR GIRDER HAS BEEN CHECKED FOR STRENGTH, DEFLECTION, AND OVERTURNING USING THE CONSTRUCTION LOADS SHOWN BELOW. CANTILEVER OVERHANG BRACKETS WERE ASSUMED FOR THE SUPPORT OF THE DECK OVERHANG PAST THE EDGE OF THE EXTERIOR GIRDER. FINISHING MACHINE WAS ASSUMED TO BE SUPPORTED 6 IN. OUTSIDE THE VERTICAL COPING FORM. THE BOTTOM OVERHANG BRACKETS WERE ASSUMED TO BE BRACED AGAINST THE INTERSECTION OF THE GIRDER BOTTOM FLANGE AND WEB.

DECK FALSEWORK LOADS
DESIGNED FOR 15 PSF FOR PERMANENT METAL STAY-IN-PLACE DECK FORMS, REMOVABLE DECK FORMS, AND 2 FT EXTERIOR WALKWAY.

CONSTRUCTION LIVE LOAD
DESIGNED FOR 20 PSF EXTENDING 2 FT PAST THE EDGE OF THE COPING AND 75 LB/FT VERTICAL FORCE APPLIED AT A DISTANCE OF 6 IN. OUTSIDE THE FACE OF COPING OVER A 30 FT LENGTH OF THE DECK CENTERED WITH THE FINISHING MACHINE.

FINISHING-MACHINE LOAD
4500 LB DISTRIBUTED OVER 10 FT ALONG THE COPING

WIND LOAD
DESIGNED FOR 70 MPH HORIZONTAL WIND LOADING IN ACCORDANCE WITH LRFD 3.8.1.

CONTINUOUS COMPOSITE
STEEL BEAM BRIDGE
3 SPANS: 49'-9", 50'-0", 49'-9"
SKEW: 15°00'00" RT.
43'-6" CLEAR ROADWAY
SR 44 OVER MUD CREEK
RUSH COUNTY

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ 08/22/2017
DESIGN ENGINEER DATE

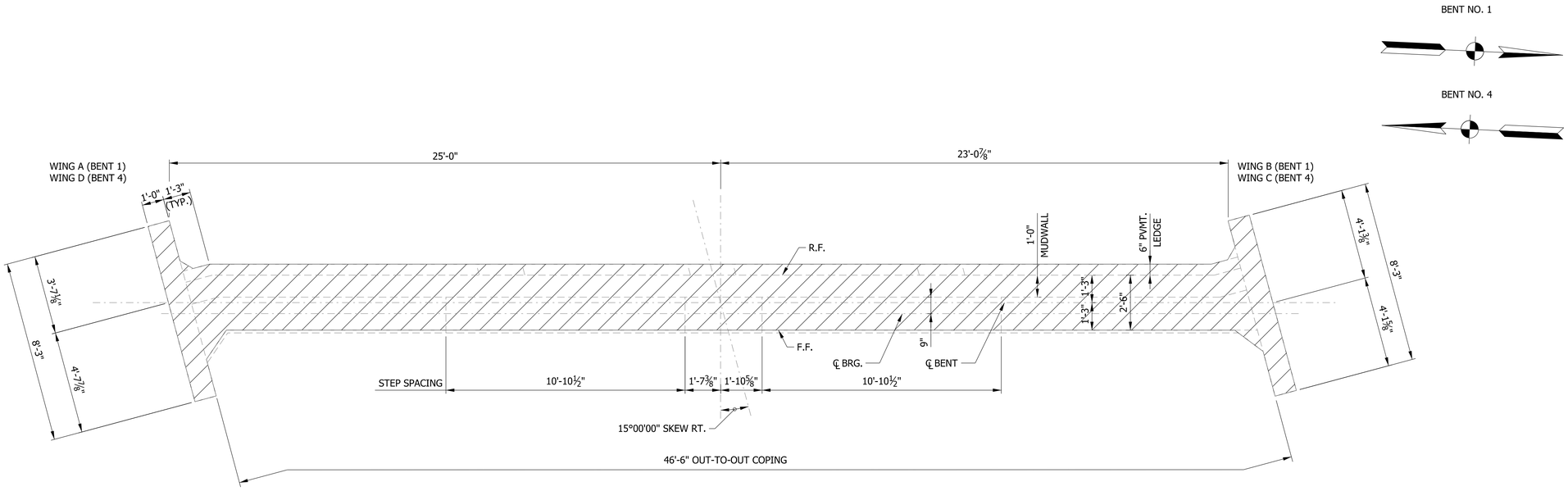
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CHECKED: SRPC 08/2016 CHECKED: SRPC 08/2016

INDIANA
DEPARTMENT OF TRANSPORTATION

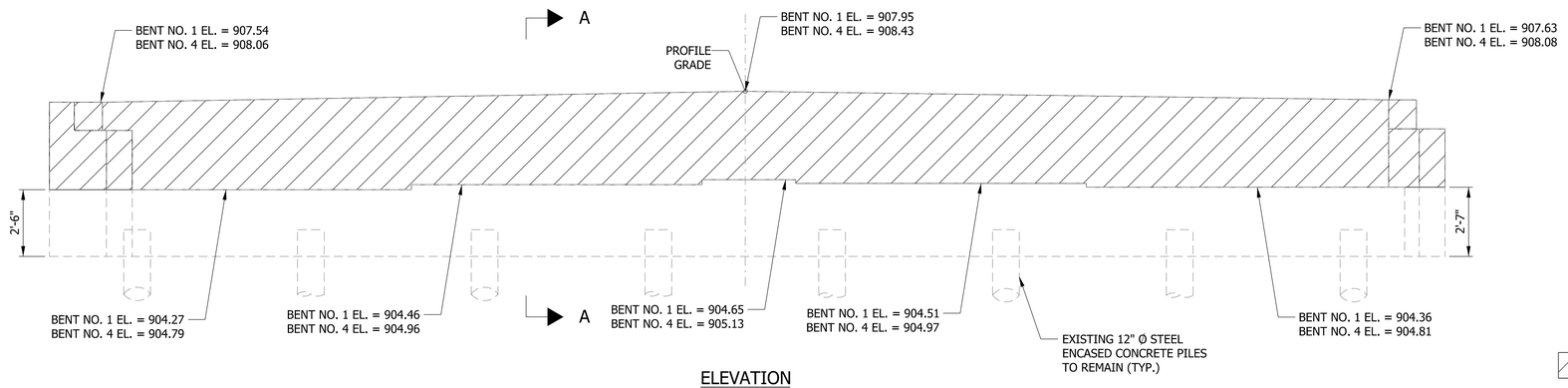
GENERAL PLAN
TYPICAL SECTIONS

SCALE	BRIDGE FILE
AS NOTED	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS NOTED	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	11 of 34
CONTRACT	PROJECT
B-38321	1500342

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PLAN
SCALE ⅜" = 1'-0"
BENT NO. 1 SHOWN, BENT NO. 4 SIMILAR



ELEVATION
SCALE ⅜" = 1'-0"
BENT NO. 1 SHOWN, BENT NO. 4 SIMILAR

LEGEND

F.F. FRONT FACE

R.F. REAR FACE

INDICATES LIMITS OF REMOVAL

NOTES

1. FOR SECTION A-A, SEE SHEET 13
2. CONCRETE DIMENSIONS ARE APPROXIMATE AND ARE FROM EXISTING BRIDGE PLANS. EXISTING ELEVATIONS SHOWN ARE BASED ON ELEVATIONS SHOWN IN PLAN SET 44-70-6634, DATED MAY 16, 1980 AND LOWERED BY 0.57' TO MORE CLOSELY MATCH THE CURRENT SURVEY.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER 08/22/2017 DATE

DESIGNED: MRS 06/2017 DRAWN: MRS 06/2017

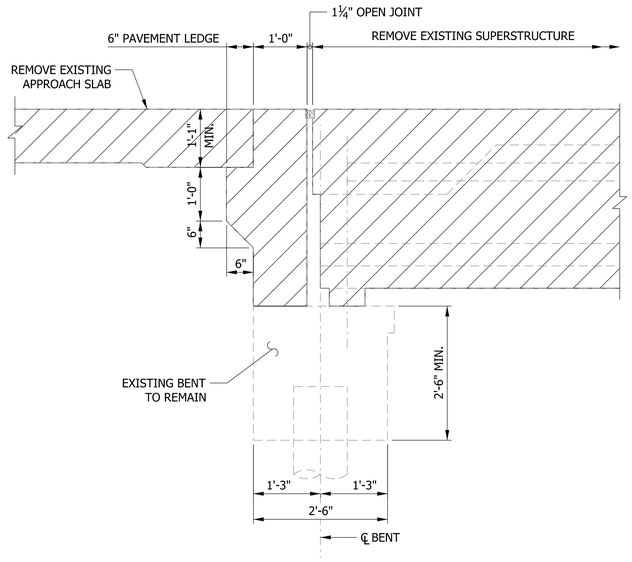
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INDIANA
DEPARTMENT OF TRANSPORTATION

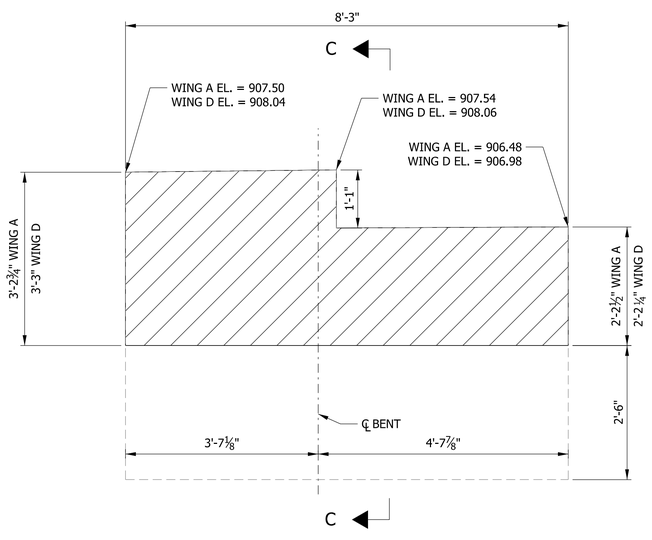
REMOVAL DETAILS

SCALE	BRIDGE FILE
⅜" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
⅜" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	12 of 34
CONTRACT	PROJECT
B-38321	1500342

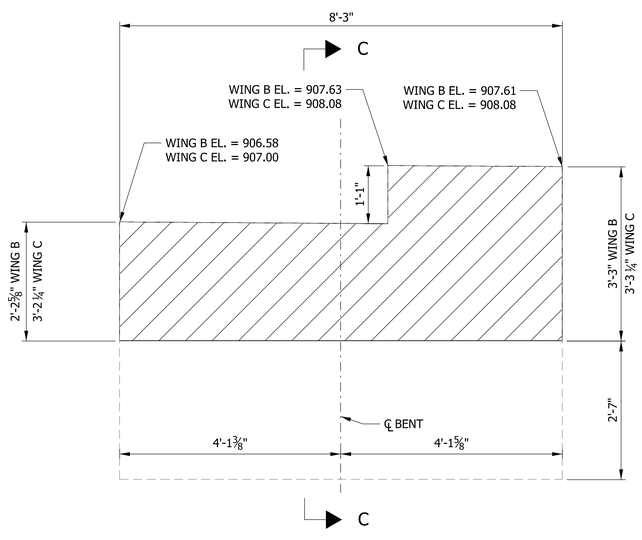
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SECTION A-A
SCALE: ¾"=1'-0"



WING A AND WING D ELEVATION
SCALE: ¾"=1'-0"



WING B AND WING C ELEVATION
SCALE: ¾"=1'-0"

SECTION C-C
SCALE: ¾"=1'-0"

NOTES

1. FOR SECTIONS A-A, SEE SHEET 12.
2. CONCRETE DIMENSIONS ARE APPROXIMATE AND ARE FROM EXISTING BRIDGE PLANS. EXISTING ELEVATIONS SHOWN ARE BASED ON ELEVATIONS SHOWN IN PLAN SET 44-70-6634, DATED MAY 16, 1980 AND LOWERED BY 0.57' TO MORE CLOSELY MATCH THE CURRENT SURVEY.

INDICATES LIMITS OF REMOVAL

SKYLER R.P. COMBS — REGISTERED — No. 11200658 — STATE OF INDIANA — PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MRS 06/2017 | DRAWN: MRS 06/2017

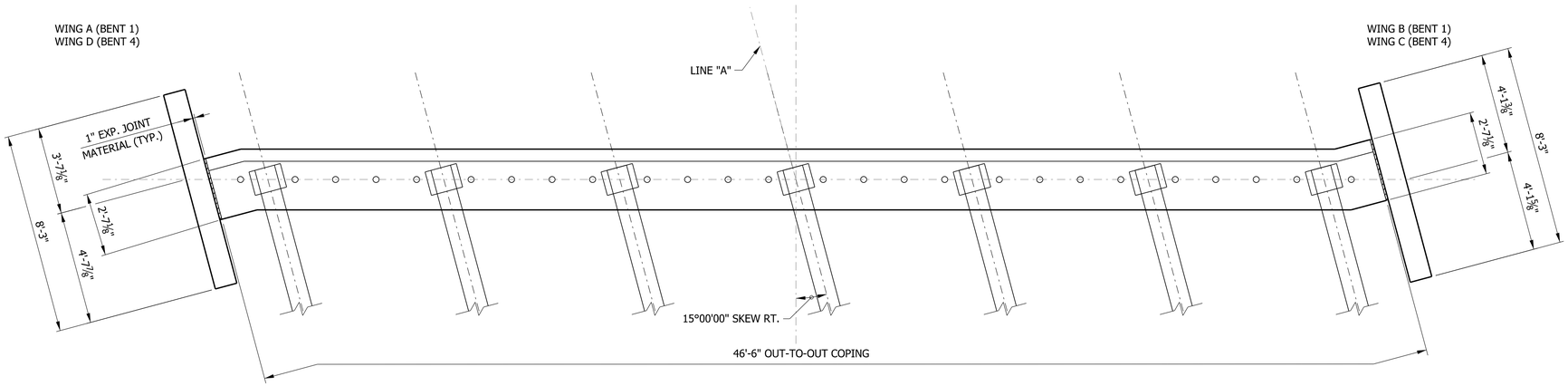
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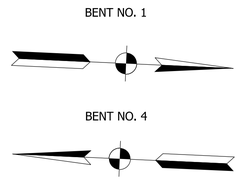
INDIANA
DEPARTMENT OF TRANSPORTATION

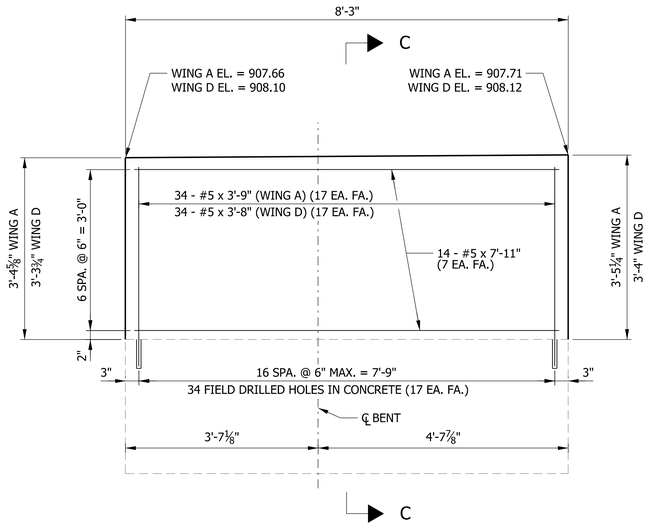
REMOVAL DETAILS

SCALE	BRIDGE FILE
⅜" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
⅜" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	13 of 34
CONTRACT	PROJECT
B-38321	1500342

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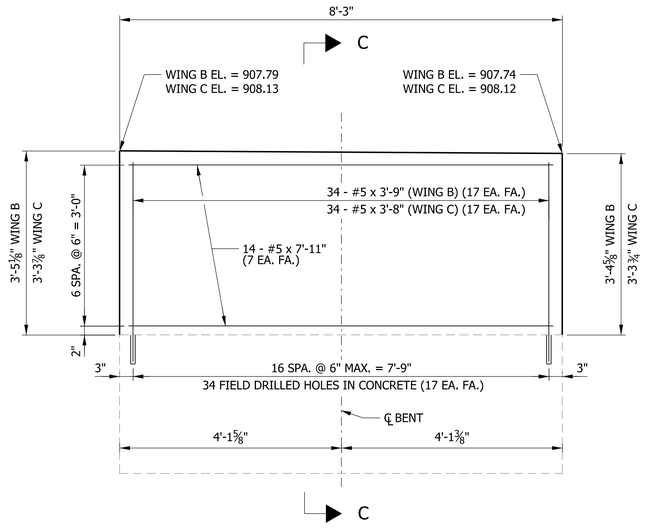






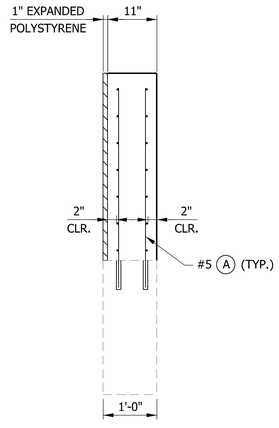
WING A AND WING D ELEVATION

SCALE: ¾"=1'-0"



WING B AND WING C ELEVATION

SCALE: ¾"=1'-0"



SECTION C-C

SCALE: ¾"=1'-0"

LEGEND

(A) BARS PLACED IN FIELD DRILLED HOLE IN CONCRETE. EMBED BAR 6" MIN. WITH AN APPROVED ANCHOR SYSTEM. MINIMUM PULLOUT = 18.6 KIP

NOTES:

1. ALL REINFORCING BARS SHALL BE EPOXY COATED.
2. FOR BILL OF MATERIALS, SEE SHEET 15.

SKYLER R.P. COMBS — REGISTERED — No. 11200658 — STATE OF INDIANA — PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MRS 06/2017 DRAWN: MRS 06/2017

CHECKED: SRPC 06/2017 CHECKED: SRPC 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

END BENT DETAILS

SCALE	BRIDGE FILE
⅜" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
⅜" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	14 of 34
CONTRACT	PROJECT
B-38321	1500342

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BILL OF MATERIALS
WINGWALL A

EPOXY COATED REINFORCING BARS

MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
#5	14	7'-11"	
#5	34	3'-9"	
TOTAL #5			249
TOTAL EPOXY-COATED REINFORCING BARS			249

CONCRETE

CONCRETE, C, SUBSTRUCTURE	1.0 CYS

MISCELLANEOUS

FIELD DRILLED HOLES IN CONCRETE	34 EACH
SURFACE SEAL (75 SFT)	1 LS

BILL OF MATERIALS
WINGWALL B

EPOXY COATED REINFORCING BARS

MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
#5	14	7'-11"	
#5	34	3'-9"	
TOTAL #5			249
TOTAL EPOXY-COATED REINFORCING BARS			249

CONCRETE

CONCRETE, C, SUBSTRUCTURE	1.0 CYS

MISCELLANEOUS

FIELD DRILLED HOLES IN CONCRETE	34 EACH
SURFACE SEAL (75 SFT)	1 LS

BILL OF MATERIALS
WINGWALL C

EPOXY COATED REINFORCING BARS

MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
#5	14	7'-11"	
#5	34	3'-8"	
TOTAL #5			246
TOTAL EPOXY-COATED REINFORCING BARS			246

CONCRETE

CONCRETE, C, SUBSTRUCTURE	0.9 CYS

MISCELLANEOUS

FIELD DRILLED HOLES IN CONCRETE	34 EACH
SURFACE SEAL (75 SFT)	1 LS

BILL OF MATERIALS
WINGWALL D

EPOXY COATED REINFORCING BARS

MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
#5	14	7'-11"	
#5	34	3'-8"	
TOTAL #5			246
TOTAL EPOXY-COATED REINFORCING BARS			246

CONCRETE

CONCRETE, C, SUBSTRUCTURE	0.9 CYS

MISCELLANEOUS

FIELD DRILLED HOLES IN CONCRETE	34 EACH
SURFACE SEAL (75 SFT)	1 LS

NOTES

1. FOR REINFORCING BAR NOTES, SEE STD. DWG. E 703-BRST-01

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ 08/22/2017
DESIGN ENGINEER DATE

DESIGNED: MRS 06/2017 DRAWN: MRS 06/2017

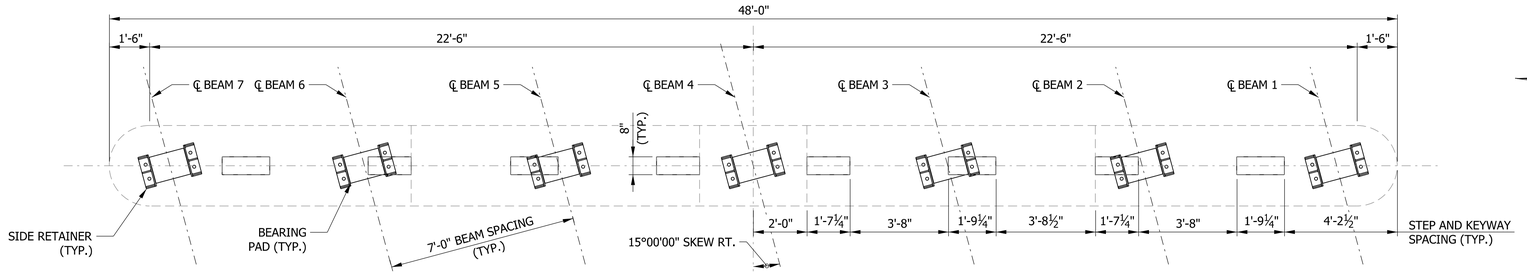
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INDIANA
DEPARTMENT OF TRANSPORTATION

END BENT DETAILS

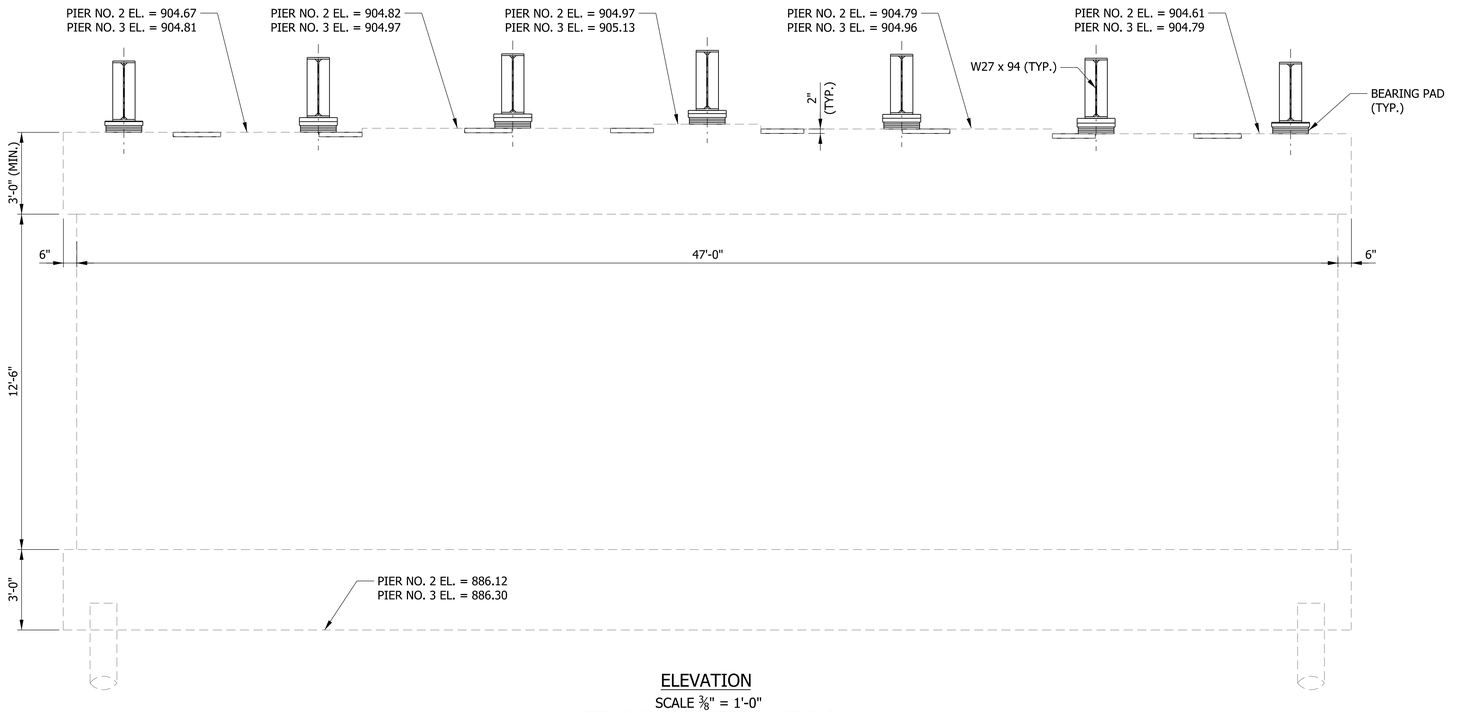
SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	15 of 34
CONTRACT	PROJECT
B-38321	1500342

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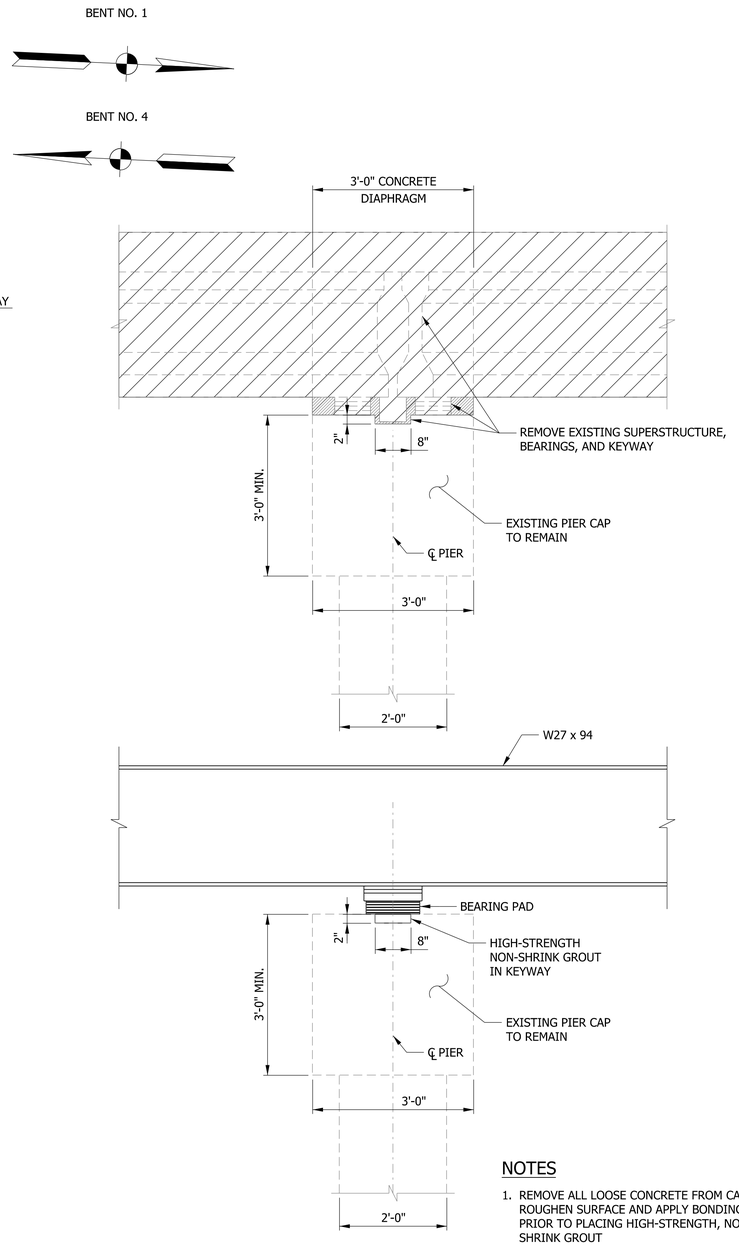
CAP PLAN

SCALE ⅜" = 1'-0"
PIER NO. 2 SHOWN, PIER NO. 3 SIMILAR



ELEVATION

SCALE ⅜" = 1'-0"
PIER NO. 2 SHOWN, PIER NO. 3 SIMILAR



NOTES

1. REMOVE ALL LOOSE CONCRETE FROM CAPS, ROUGHEN SURFACE AND APPLY BONDING AGENT PRIOR TO PLACING HIGH-STRENGTH, NON-SHRINK GROUT
2. CONCRETE DIMENSIONS ARE APPROXIMATE AND ARE FROM EXISTING BRIDGE PLANS.

INDICATES LIMITS OF REMOVAL

SKYLER R.P. COMBS — REGISTERED — No. 11200658 — STATE OF INDIANA — PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER — 08/22/2017 DATE

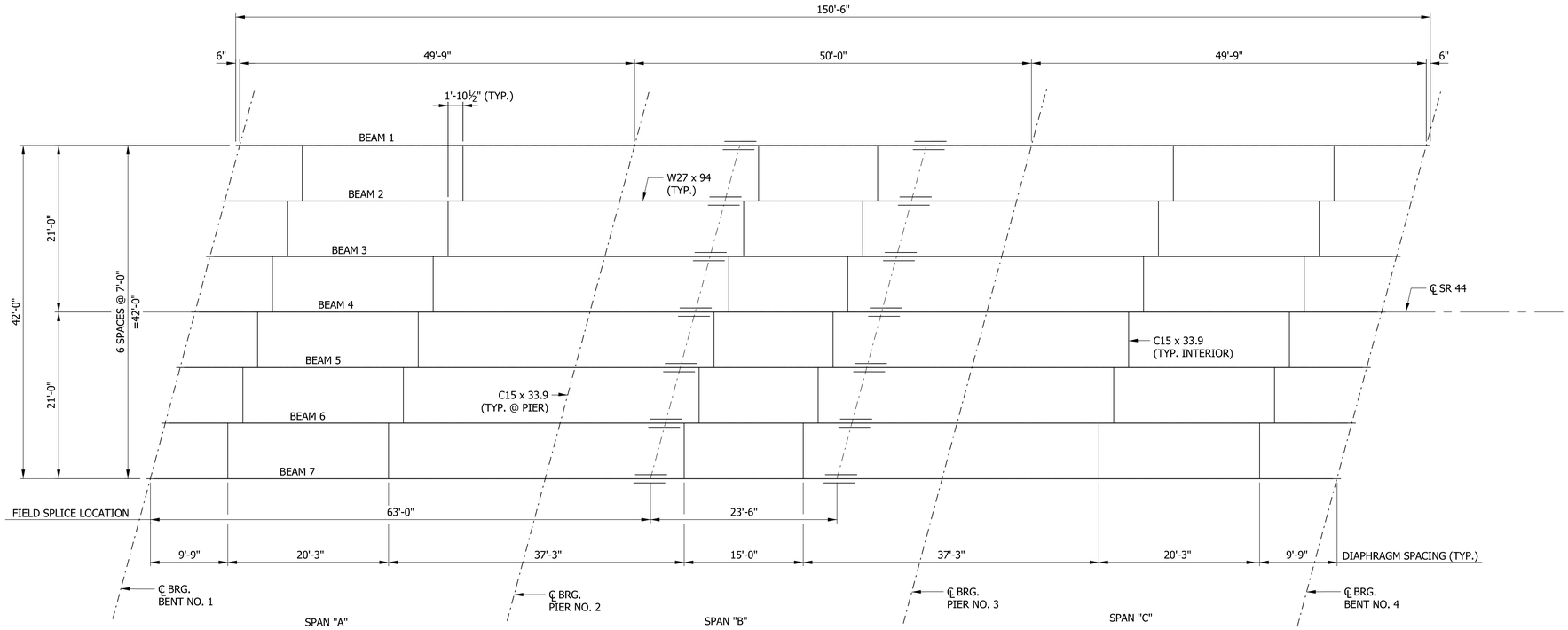
DESIGNED: MRS 06/2017 | DRAWN: MRS 06/2017
CHECKED: SRS 06/2017 | CHECKED: SRS 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

PIER DETAILS

SCALE	BRIDGE FILE
⅜" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
⅜" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	16 of 34
CONTRACT	PROJECT
B-38321	1500342

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model: Pier no. 2 details
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FRAMING PLAN

FABRICATION AND ERECTION NOTES:

1. ALL STRUCTURAL STEEL, UNLESS NOTED, IS TO BE ASTM A709, GRADE 50W.
2. ALL BOLTS SHALL BE HIGH STRENGTH A325 TYPE 3 BOLTS, ⅞" DIAM. UNLESS NOTED. OPEN HOLES SHALL BE 15/16" DIAM. UNLESS NOTED.
3. THE DIMENSIONS FOR THESE DETAIL PLANS ARE BASED ON CONSTRUCTION PLANS AND/OR SHOP DRAWINGS OF THE ORIGINAL STRUCTURE AND SURVEY. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY CONTROLLING DIMENSIONS IN THE FIELD PRIOR TO FABRICATION.
4. THE WEIGHT OF HIGH STRENGTH BOLTS IS NOT INCLUDED IN THE ESTIMATED WEIGHT OF THE STRUCTURAL STEEL.

   ESTIMATED WEIGHT OF STRUCTURAL STEEL: 117,950 LBS. (TOTAL)

SKYLER R.P. COMBS REGISTERED NO. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MEW 08/2016 | DRAWN: DCK 05/2017

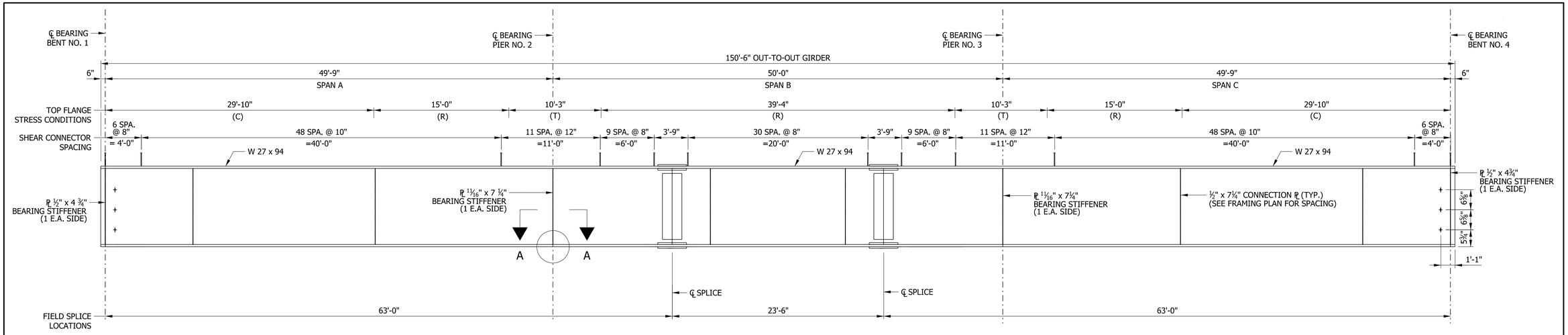
CHECKED: SRPC 08/2016 | CHECKED: MRS 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

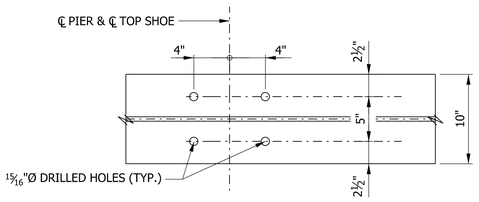
FRAMING PLAN

SCALE	BRIDGE FILE
1/8" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
1/8" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	17 of 34
CONTRACT	PROJECT
B-38321	1500342

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file: \\indot\00\289projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-frp01.dgn

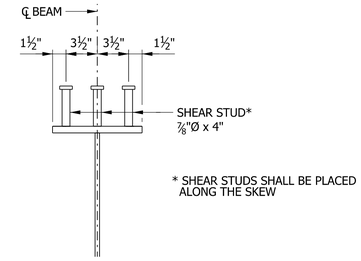


INTERIOR GIRDER ELEVATION



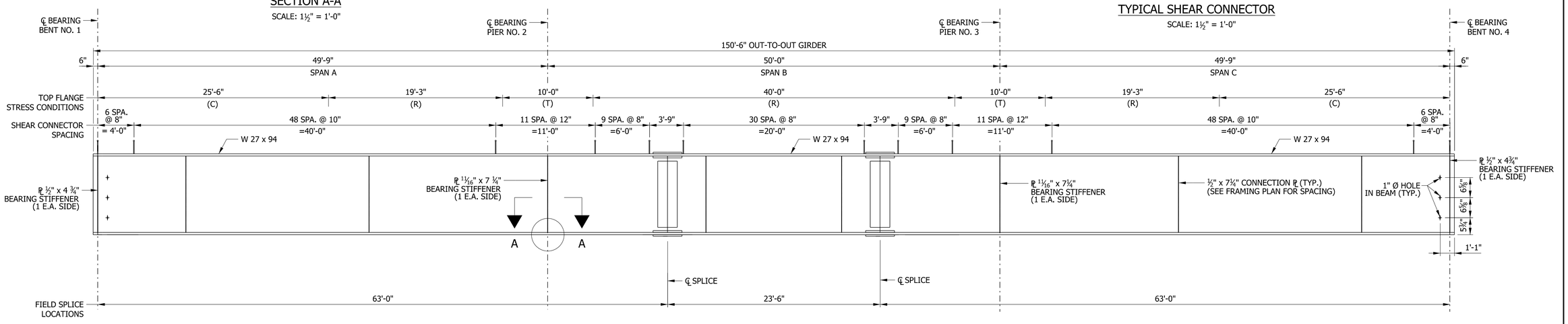
SECTION A-A

SCALE: 1½" = 1'-0"



TYPICAL SHEAR CONNECTOR

SCALE: 1½" = 1'-0"



EXTERIOR GIRDER ELEVATION

LEGEND
(T) - TENSION
(R) - REVERSAL
(C) - COMPRESSION

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: DCK 06/2017 DRAWN: DCK 06/2017

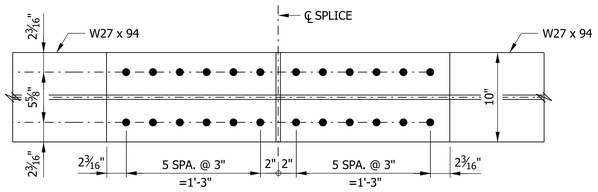
CHECKED: MRS 06/2017 CHECKED: MRS 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

STRUCTURAL STEEL DETAILS

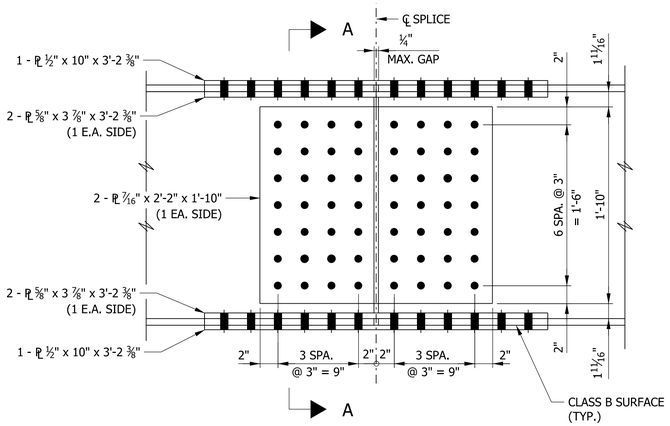
SCALE	BRIDGE FILE
3/16" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
3/4" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	18 of 34
CONTRACT	PROJECT
B-38321	1500342

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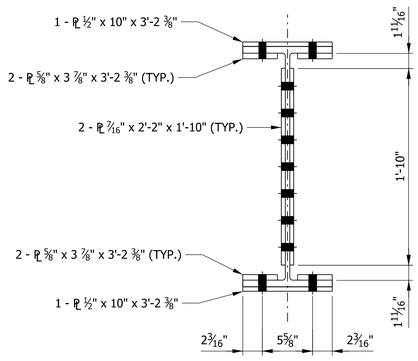
TOP FLANGE PLAN

SCALE: 1½" = 1'-0"



TYPICAL SPLICE DETAIL

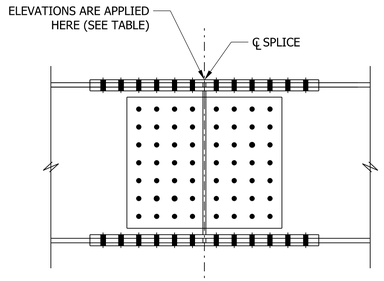
SCALE: 1½" = 1'-0"



SECTION A-A

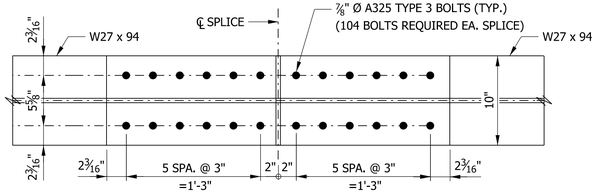
SCALE: 1½" = 1'-0"

TABLE OF ELEVATIONS FOR TOP OF BEAM SPLICE		
BEAM	SPLICE 2-1	SPLICE 2-2
1	907.408	907.464
2	907.542	907.600
3	907.676	907.737
4	907.810	907.873
5	907.664	907.729
6	907.517	907.585
7	907.371	907.441



TYPICAL SPLICE DETAIL

NOT TO SCALE

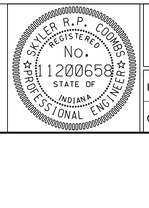


BOTTOM FLANGE PLAN

SCALE: 1½" = 1'-0"

DIAPHRAGM NOTES:

1. ALL STRUCTURAL STEEL, UNLESS NOTED, IS TO BE ASTM A709, GRADE 50W
2. ALL BOLTS SHALL BE HIGH STRENGTH A325 TYPE 3 BOLTS, ⅞" DIAM. UNLESS NOTED. OPEN HOLES SHALL BE 15/16" DIAM. UNLESS NOTED.



RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: DCK 06/2017 DRAWN: DCK 06/2017

CHECKED: MRS 06/2017 CHECKED: MRS 06/2017

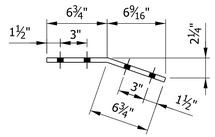
INDIANA DEPARTMENT OF TRANSPORTATION

STRUCTURAL STEEL DETAILS

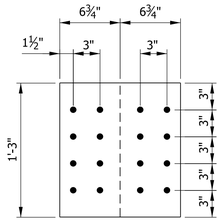
SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342

SURVEY BOOK	SHEETS
ELECTRONIC	19 of 34
CONTRACT	PROJECT
B-38321	1500342

mstocker
8/22/2017 3:21:00 pm
model: Sheet 1
file: \\indw001269\projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-bmd02.dgn



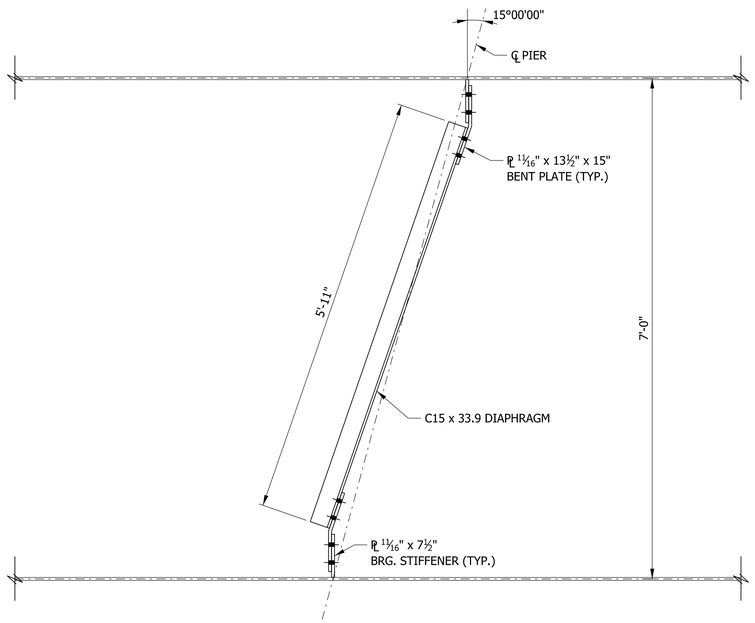
PLAN



ELEVATION

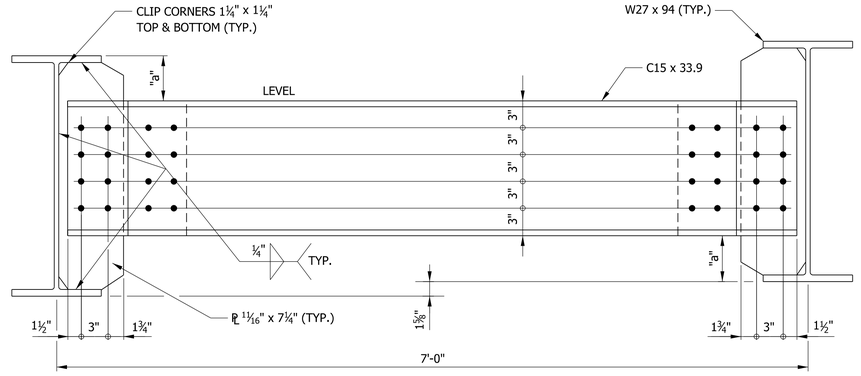
BENT PLATE DETAILS

SCALE: 1½" = 1'-0"



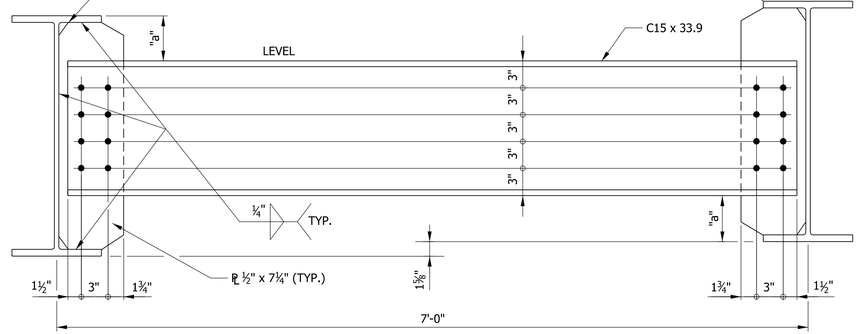
PLAN - PIER DIAPHRAGM

SCALE: 1" = 1'-0"



ELEVATION - PIER DIAPHRAGM

SCALE: 1½" = 1'-0"



TYPICAL DIAPHRAGM DETAIL

SCALE: 1½" = 1'-0"

DIAPHRAGM NOTES:

1. ALL STRUCTURAL STEEL, UNLESS NOTED, IS TO BE ASTM A709, GRADE 50W
2. ALL BOLTS SHALL BE HIGH STRENGTH A325 TYPE 3 BOLTS, ⅞" DIAM. UNLESS NOTED. OPEN HOLES SHALL BE 15⁄16" DIAM. UNLESS NOTED.

SKYLER R.P. COOMBS — REGISTERED — No. 11200658 — STATE OF INDIANA — PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: DCK 06/2017 | DRAWN: DCK 06/2017

CHECKED: MRS 06/2017 | CHECKED: MRS 06/2017

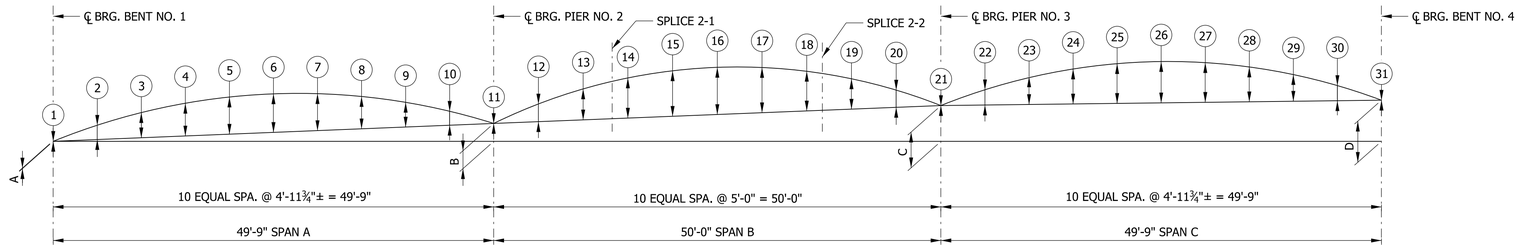
INDIANA
DEPARTMENT OF TRANSPORTATION

STRUCTURAL STEEL DETAILS

SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342

SURVEY BOOK	SHEETS
ELECTRONIC	20 of 34
CONTRACT	PROJECT
B-38321	1500342

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model: Sheet 2
file: \\indw001289\projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-bmd02.dgn



NO LOAD CAMBER DIAGRAM
NOT TO SCALE

BLOCKING DIMENSIONS (IN)				
BEAM NO	A	B	C	D
1	0.00	3.05	4.45	4.19
2	0.00	3.11	4.58	4.38
3	0.00	3.17	4.70	4.57
4	0.00	3.23	4.82	4.75
5	0.00	3.30	4.95	4.94
6	0.00	3.36	5.07	5.13
7	0.00	3.42	5.20	5.31

TABLE OF CAMBERS (INCHES) - BEAM 1 AND 7

	SPAN "A"											SPAN "B"												SPAN "C"									
LOCATION	1	2	3	4	5	6	7	8	9	10	11	12	13	Splice 2-1	14	15	16	17	18	Splice 2-2	19	20	21	22	23	24	25	26	27	28	29	30	31
DEAD LOAD - STEEL	0.00	0.03	0.05	0.07	0.08	0.08	0.07	0.05	0.03	0.01	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.01	0.03	0.05	0.07	0.08	0.08	0.07	0.05	0.03	0.00
DEAD LOAD - SLAB AND FORMS	0.00	0.17	0.32	0.42	0.48	0.48	0.42	0.33	0.21	0.09	0.00	-0.03	-0.02	0.00	0.01	0.03	0.04	0.03	0.01	0.00	-0.02	-0.03	0.00	0.09	0.21	0.33	0.42	0.48	0.48	0.42	0.32	0.17	0.00
DEAD LOAD - RAILING	0.00	0.03	0.05	0.07	0.08	0.08	0.07	0.05	0.03	0.01	0.00	-0.01	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.00	0.00	-0.01	0.00	0.01	0.03	0.05	0.07	0.08	0.08	0.07	0.05	0.03	0.00
SUBTOTAL - DEAD LOAD	0.00	0.23	0.42	0.56	0.64	0.64	0.56	0.43	0.27	0.11	0.00	-0.04	-0.02	0.00	0.01	0.05	0.06	0.05	0.01	0.00	-0.02	-0.04	0.00	0.11	0.27	0.43	0.56	0.64	0.64	0.56	0.42	0.23	0.00
GEOMETRIC CAMBER	0.00	0.07	0.13	0.17	0.20	0.21	0.20	0.17	0.13	0.07	0.00	0.07	0.13	0.16	0.17	0.20	0.21	0.20	0.17	0.17	0.13	0.07	0.00	0.07	0.13	0.17	0.20	0.21	0.20	0.17	0.13	0.07	0.00
TOTAL CAMBER	0.00	0.30	0.55	0.73	0.84	0.85	0.76	0.60	0.40	0.18	0.00	0.03	0.11	0.16	0.18	0.25	0.27	0.25	0.18	0.17	0.11	0.03	0.00	0.18	0.40	0.60	0.76	0.85	0.84	0.73	0.55	0.30	0.00

TABLE OF CAMBERS (INCHES) - BEAM 2, 3, 4, 5, AND 6

	SPAN "A"											SPAN "B"												SPAN "C"									
LOCATION	1	2	3	4	5	6	7	8	9	10	11	12	13	Splice 2-1	14	15	16	17	18	Splice 2-2	19	20	21	22	23	24	25	26	27	28	29	30	31
DEAD LOAD - STEEL	0.00	0.03	0.06	0.07	0.08	0.08	0.07	0.06	0.04	0.02	0.00	-0.01	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.00	0.00	-0.01	0.00	0.02	0.04	0.06	0.07	0.08	0.08	0.07	0.06	0.03	0.00
DEAD LOAD - SLAB AND FORMS	0.00	0.22	0.41	0.54	0.61	0.60	0.54	0.41	0.26	0.11	0.00	-0.04	-0.02	0.00	0.01	0.04	0.05	0.04	0.01	0.00	-0.02	-0.04	0.00	0.11	0.26	0.41	0.54	0.60	0.61	0.54	0.41	0.22	0.00
DEAD LOAD - RAILING	0.00	0.02	0.03	0.04	0.05	0.05	0.04	0.03	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.03	0.04	0.05	0.05	0.04	0.03	0.02	0.00
SUBTOTAL - DEAD LOAD	0.00	0.27	0.50	0.65	0.74	0.73	0.65	0.50	0.32	0.14	0.00	-0.05	-0.02	0.00	0.01	0.05	0.06	0.05	0.01	0.00	-0.02	-0.05	0.00	0.14	0.32	0.50	0.65	0.73	0.74	0.65	0.50	0.27	0.00
GEOMETRIC CAMBER	0.00	0.07	0.13	0.17	0.20	0.21	0.20	0.17	0.13	0.07	0.00	0.07	0.13	0.16	0.17	0.20	0.21	0.20	0.17	0.17	0.13	0.07	0.00	0.07	0.13	0.17	0.20	0.21	0.20	0.17	0.13	0.07	0.00
TOTAL CAMBER	0.00	0.34	0.63	0.82	0.94	0.94	0.85	0.67	0.45	0.21	0.00	0.02	0.11	0.16	0.18	0.25	0.27	0.25	0.18	0.17	0.11	0.02	0.00	0.21	0.45	0.67	0.85	0.94	0.94	0.82	0.63	0.34	0.00

NOTE:

1. CAMBER INCLUDES DEAD LOAD OF GIRDER, SLAB, FORMS, AND BARRIERS AS WELL AS CURVE GEOMETRY. FUTURE WEARING SURFACE IS NOT INCLUDED.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MRS 06/2017 DRAWN: MRS 06/2017

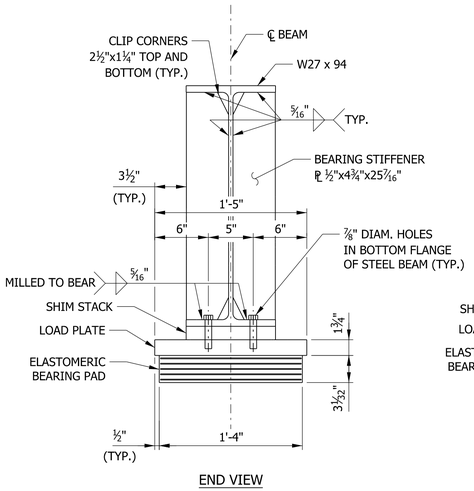
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INDIANA
DEPARTMENT OF TRANSPORTATION

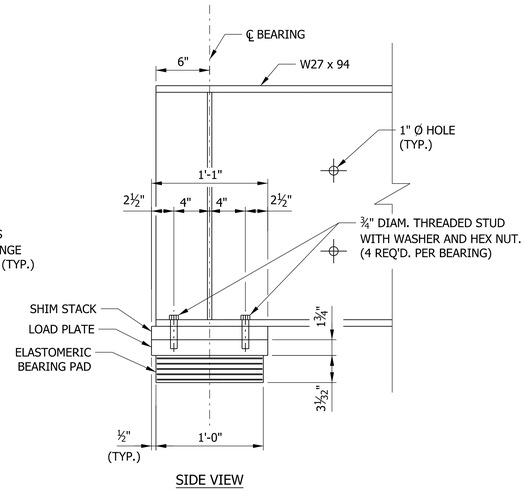
STRUCTURAL STEEL DETAILS

SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	21 of 34
CONTRACT	PROJECT
B-38321	1500342

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8/22/2017 3:21:01 pm
model: Camber details
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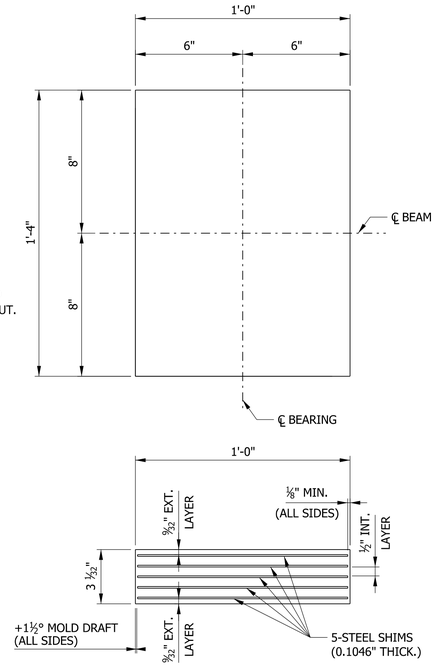
END VIEW



SIDE VIEW

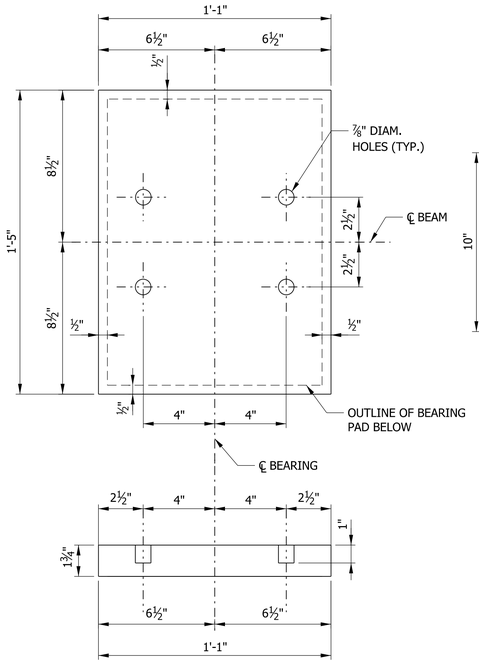
END BENT BEARING DETAILS

(14 REQUIRED)
SCALE: 1½" = 1'-0"



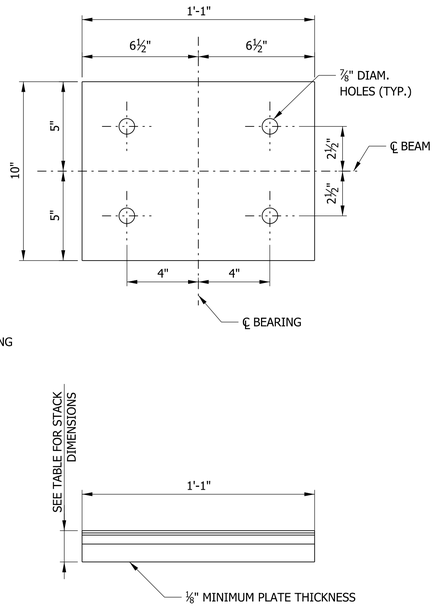
ELASTOMERIC BEARING PAD DETAILS

SCALE: 3" = 1'-0"



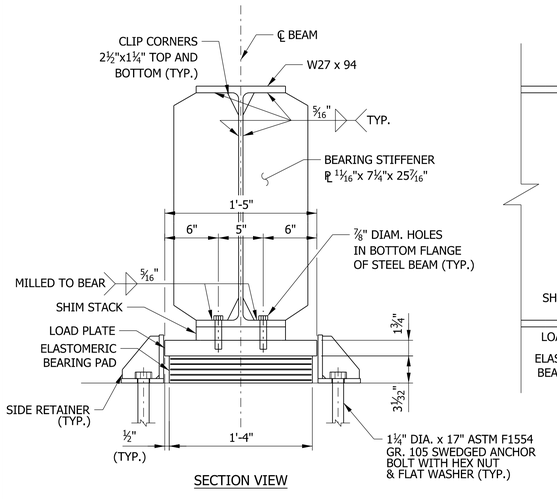
LOAD PLATE DETAILS

SCALE: 3" = 1'-0"

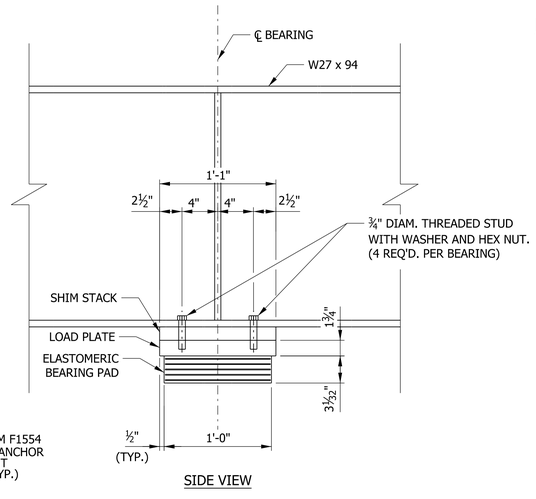


SHIM STACK DETAILS

SCALE: 3" = 1'-0"



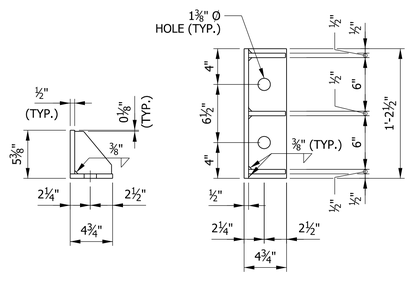
SECTION VIEW



SIDE VIEW

PIER BEARING DETAILS

(14 REQUIRED)
SCALE: 1½" = 1'-0"



SIDE RETAINER DETAILS

SCALE: 1½" = 1'-0"

BEARING NOTES:

1. FABRICATE THE ELASTOMERIC BEARING PADS TO THE DESIGN AND DIMENSIONS SHOWN ON THE PLANS.
2. FINISHED BEARING SURFACES SHALL BE CLEAN AND FREE OF LOOSE MATERIAL BEFORE PLACING BEARING PADS.
3. STEEL USED IN LOAD PLATES AND SHIM STACK SHALL BE A709, GRADE 50.
4. ALL BEARING PLATES, SHIMS, BOLTS AND WASHERS SHALL BE GALVANIZED IN ACCORDANCE WITH ASTM A123 OR A153.
5. ELASTOMERIC BEARINGS SHALL BE VULCANIZED TO THE LOAD PLATE.
6. SHIM STACK DIMENSIONS SHALL BE FIELD VERIFIED PRIOR TO FABRICATION.
7. ALL ELASTOMER SHALL BE 55 DUROMETER HARDNESS ON THE SHORE A SCALE, WITH A SHEAR MODULES @ 73°F OF BETWEEN 112 AND 165 PSI.
8. THE HOLES FOR THE BEARING ANCHOR BOLTS SHALL BE DRILLED. GROUT FOR THIS SYSTEM: THE GROUT SHALL BE HIGH STRENGTH, NON-SHRINK, NON-METALLIC GROUT. PRIOR TO GROUTING THE ANCHOR BOLTS THE INSIDES OF THE HOLES SHALL BE COATED WITH A BONDING AGENT.
9. SEE STANDARD DWG. E 726-BEBP-04 & 05, BEARING TYPE S5-A, FOR ADDITIONAL DETAILS.

TABLE OF SHIM STACK DIMENSIONS

BEAM	BENT 1	PIER 2	PIER 3	BENT 4
1	¾"	⅛"	0"	0"
2	2⅜"	1¾"	1½"	1½"
3	2⅛"	1½"	1¼"	1¼"
4	2"	1⅜"	1"	1"
5	2⅜"	1⅝"	1¼"	1¼"
6	2¾"	2"	1⅝"	1½"
7	1"	¼"	0"	0"

mstocker
8/22/2017 3:21:04 pm
model: Brg and detail details
file: \\indot\dfs\projects\6e4359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\c6s64259-s-br-bd01.dgn

SKYLER R.P. COMBS
REGISTERED
No. 11200658
STATE OF INDIANA
PROFESSIONAL ENGINEER

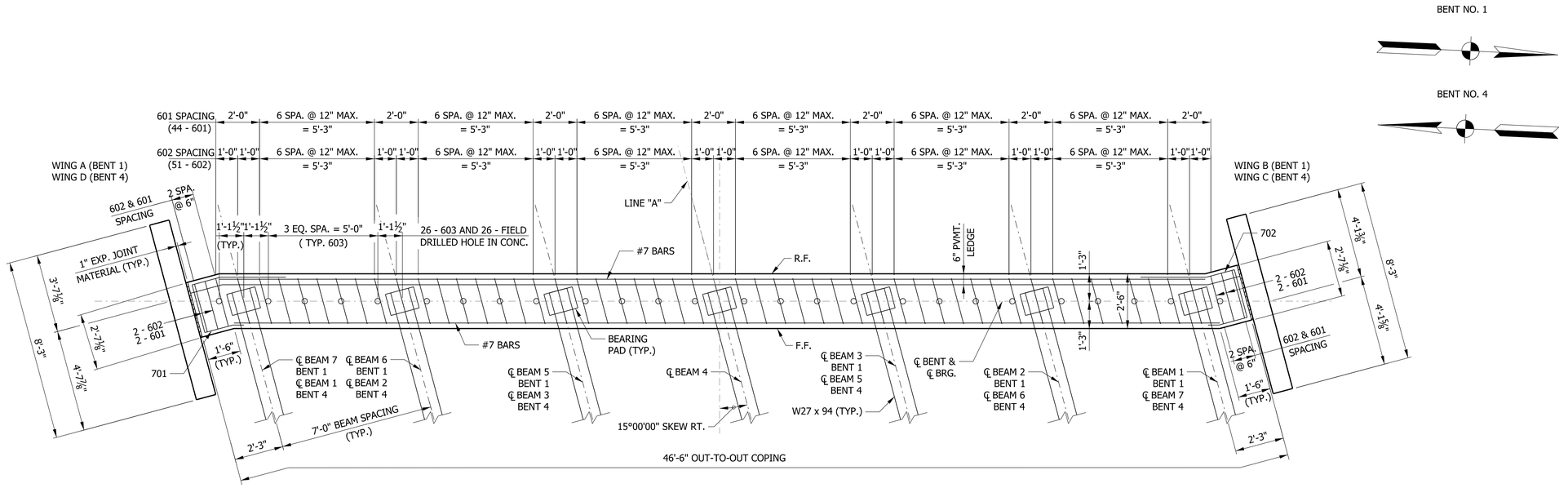
RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MPK 06/2017 | DRAWN: MPK 06/2017
CHECKED: SRPC 06/2017 | CHECKED: SRPC 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

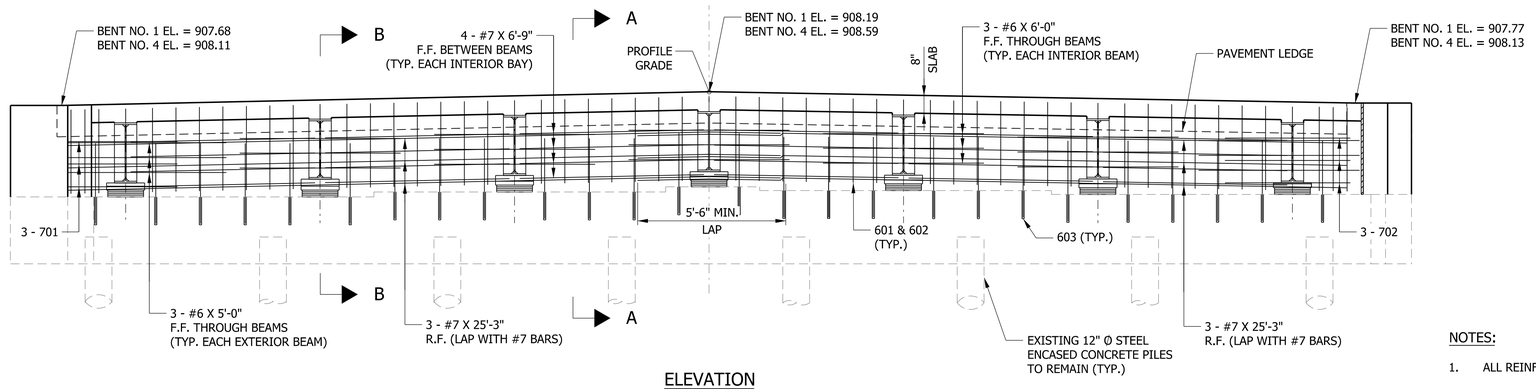
BEARING DETAILS

SCALE	BRIDGE FILE
AS NOTED	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS NOTED	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	22 of 34
CONTRACT	PROJECT
B-38321	1500342



PLAN

SCALE ⅜" = 1'-0"
BENT NO. 1 SHOWN, BENT NO. 4 SIMILAR



ELEVATION

SCALE ⅜" = 1'-0"
BENT NO. 1 SHOWN, BENT NO. 4 SIMILAR

LEGEND

F.F. FRONT FACE

R.F. REAR FACE

NOTES:

1. ALL REINFORCING BARS SHALL BE EPOXY COATED.
2. FOR REINFORCING BAR BENDING DIAGRAMS AND BILL OF MATERIALS, SEE SHEET 28.
3. FOR SECTION A-A AND B-B, SEE SHEET 24.
3. THE TOP REINFORCING IN THE DECK SHALL BE SECURELY TIED DOWN TO THE DECK FORMS AND/OR BEAMS TO PREVENT LIFTING DURING CONCRETE PLACEMENT.
4. SUITABLE RESTRAINT SHALL BE PROVIDED TO PREVENT THE ROTATION OF THE EXTERIOR BEAMS FROM CONSTRUCTION LOAD SUCH AS FINISHING MACHINE AND FORMS.
5. SEMI-INTEGRAL END BENT DIAPHRAGMS SHALL BE POURED FIRST.

SKYLER R.F. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

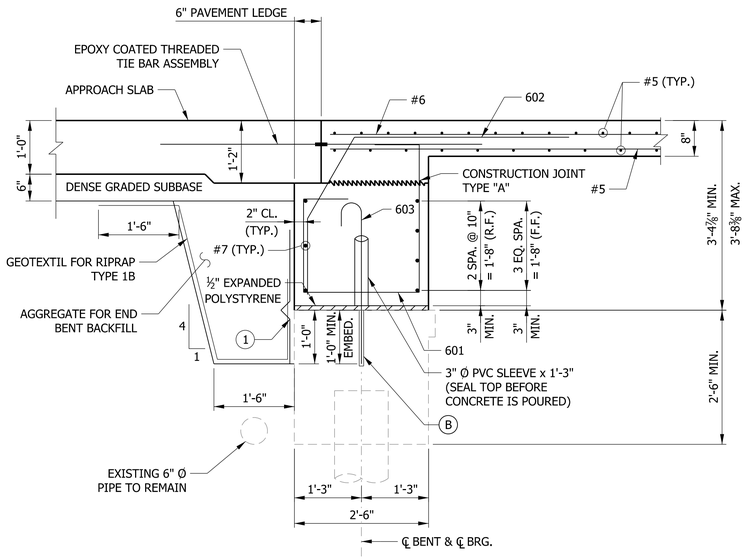
DESIGNED: MRS 06/2017 | DRAWN: MRS 06/2017
CHECKED: SRPC 06/2017 | CHECKED: SRPC 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

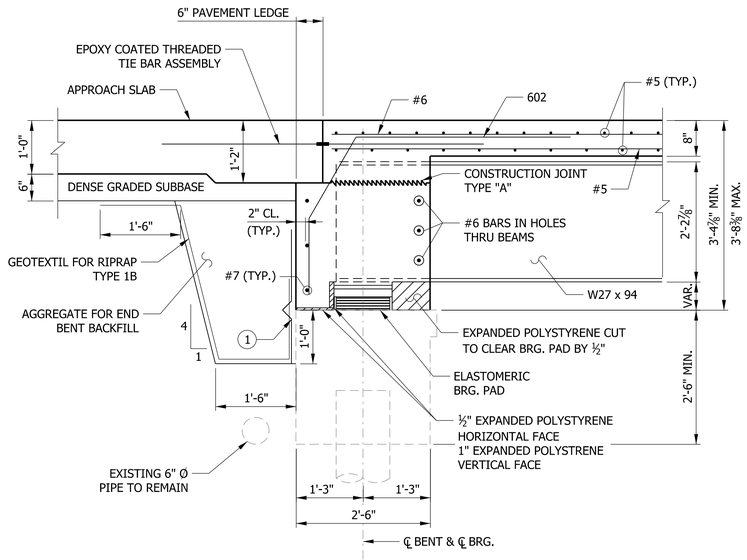
SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	23 of 34
CONTRACT	PROJECT
B-38321	1500342

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model: Bent proposed
file: I:\indot\00\projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64259-s-br-ebs001.dgn



SECTION A-A
SCALE: ¾"=1'-0"



SECTION B-B
SCALE: ¾"=1'-0"

LEGEND

(1) POLYCHLOROPRENE JOINT MEMBRANE ATTACHED TO CONCRETE

(B) BARS PLACED IN FIELD DRILLED HOLE IN CONCRETE. EMBED BAR 12" WITH AN APPROVED ANCHOR SYSTEM. MINIMUM PULLOUT = 26.5 KIP

F.F. FRONT FACE

R.F. REAR FACE

NOTES:

1. ALL REINFORCING BARS SHALL BE EPOXY COATED.
2. FOR REINFORCING BAR BENDING DIAGRAMS AND BILL OF MATERIALS, SEE SHEET 28.
3. THE TOP REINFORCING IN THE DECK SHALL BE SECURELY TIED DOWN TO THE DECK FORMS AND/OR BEAMS TO PREVENT LIFTING DURING CONCRETE PLACEMENT.
4. SUITABLE RESTRAINT SHALL BE PROVIDED TO PREVENT THE ROTATION OF THE EXTERIOR BEAMS FROM CONSTRUCTION LOAD SUCH AS FINISHING MACHINE AND FORMS.
5. SEMI-INTEGRAL END BENT DIAPHRAGMS SHALL BE POURED FIRST.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ 08/22/2017
DESIGN ENGINEER DATE

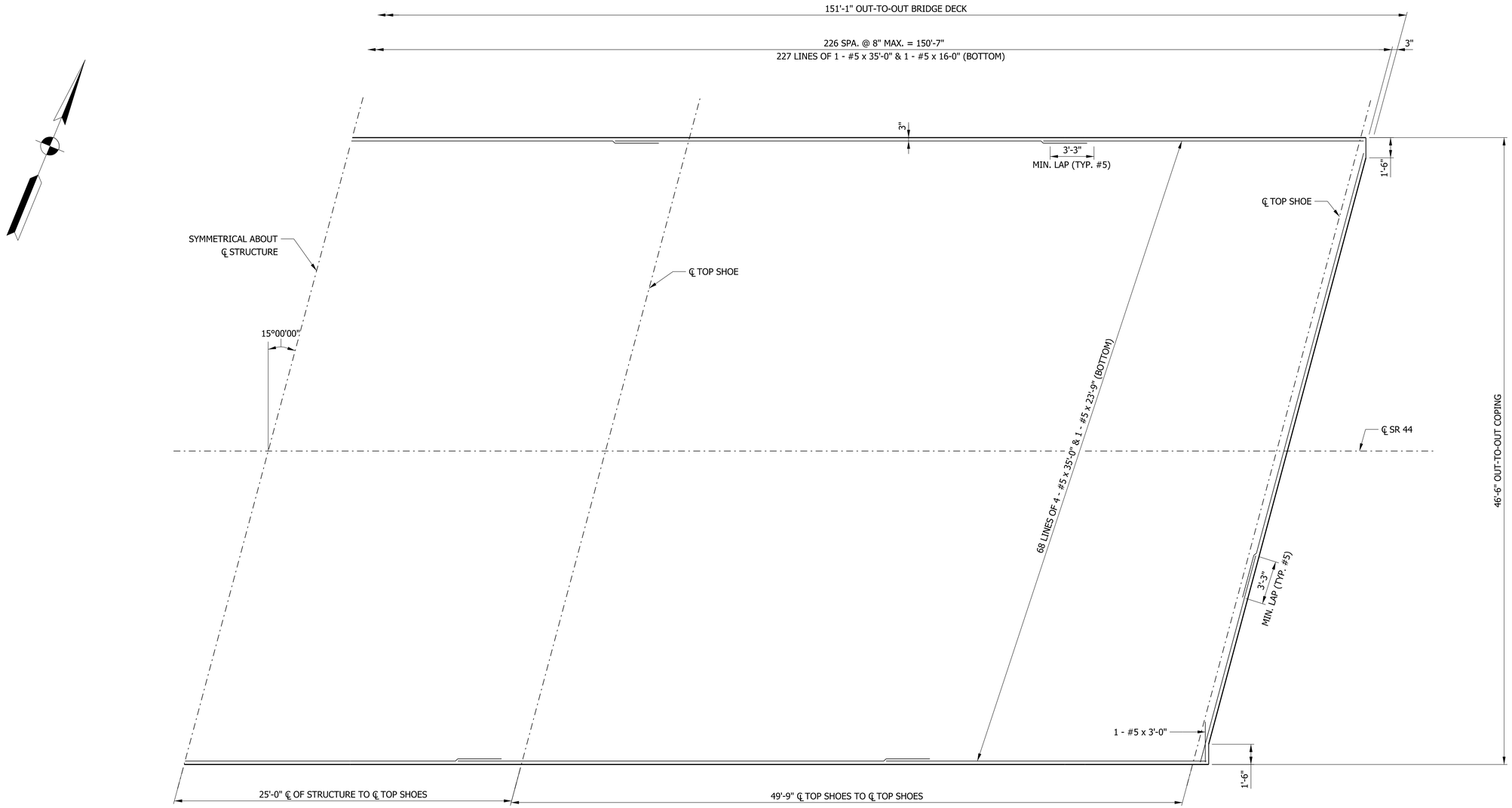
DESIGNED: MRS 06/2017 DRAWN: MRS 06/2017
CHECKED: SRPC 06/2017 CHECKED: SRPC 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
⅜" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
⅜" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	24 of 34
CONTRACT	PROJECT
B-38321	1500342

mstocker
8/22/2017 3:21:06 pm
model: Bent proposed section
file: I:\indot\05\projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64259-s-br-ebs001.dgn



HALF SLAB PLAN
SHOWING BOTTOM REINFORCEMENT

NOTES:

1. ALL REINFORCING BARS TO BE EPOXY COATED.
2. MIN. LAP #5 = 3'-3"
3. FOR REINFORCING BAR BENDING DIAGRAMS AND BILL OF MATERIALS, SEE SHEET 28.

SKYLER R.P. COOMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: DCK 06/2017 | DRAWN: DCK 06/2017

CHECKED: MRS 06/2017 | CHECKED: MRS 06/2017

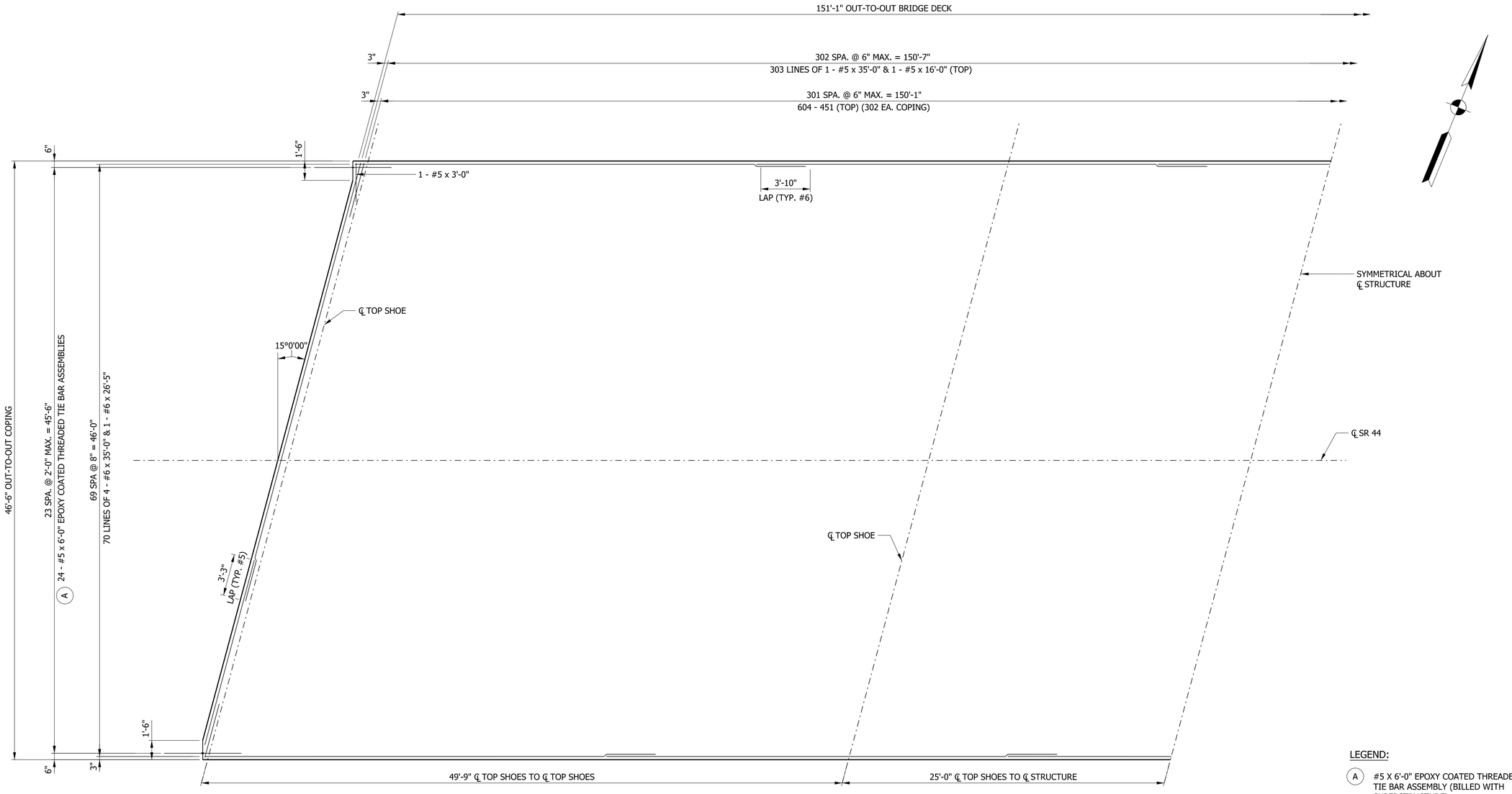
INDIANA
DEPARTMENT OF TRANSPORTATION

SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
1/4" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
1/4" = 1'-0"	1500342

SURVEY BOOK	SHEETS
ELECTRONIC	25 of 34
CONTRACT	PROJECT
B-38321	1500342

mstocker
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model: hall slab bottom reinforcement
file: \\indot\i200\projects\64359 hwks gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-ss001.dgn



HALF SLAB PLAN
SHOWING TOP REINFORCEMENT

LEGEND:

(A) #5 X 6'-0" EPOXY COATED THREADED TIE BAR ASSEMBLY (BILLED WITH SUPERSTRUCTURE)

NOTES:

1. ALL REINFORCING BARS TO BE EPOXY COATED.
2. MIN. LAP #5 = 3'-3"
3. MIN. LAP #6 = 3'-10"
4. FOR REINFORCING BAR BENDING DIAGRAMS AND BILL OF MATERIALS, SEE SHEET 28.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER 08/22/2017 DATE

DESIGNED: DCK 06/2017 DRAWN: DCK 06/2017

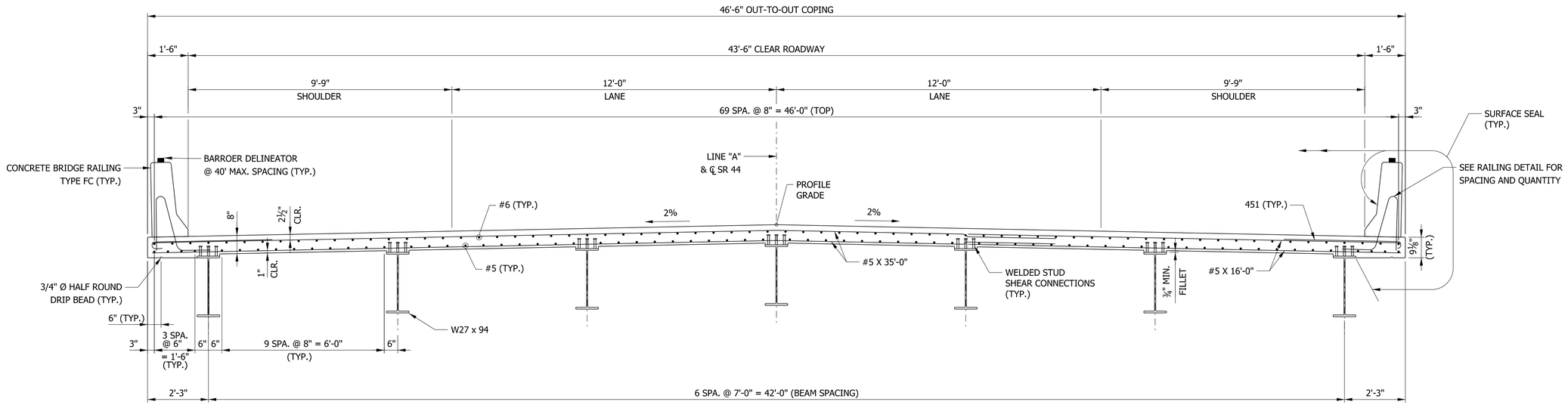
CHECKED: MRS 06/2017 CHECKED: MRS 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
1/4" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
1/4" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	26 of 34
CONTRACT	PROJECT
B-38321	1500342

mstocker
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model: Half slab top reinforcement
file: L:\indot\0_0\projects\64259 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cs\64259-s-br-ssd01.dgn



TYPICAL SECTION

NOTES:

1. ALL REINFORCING BARS SHALL BE EPOXY COATED.
2. FOR REINFORCING BAR BENDING DIAGRAMS AND BILL OF MATERIALS, SEE SHEET 28.

SKYLER R.P. COOMBS
REGISTERED
No. 11200658
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER 08/22/2017 DATE

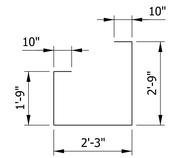
DESIGNED: DCK 06/2017
DRAWN: DCK 06/2017
CHECKED: MRS 06/2017
CHECKED: MRS 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

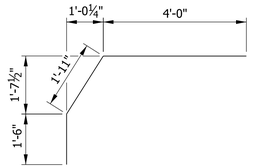
SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
1/2" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
1/2" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	27 of 34
CONTRACT	PROJECT
B-38321	1500342

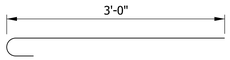
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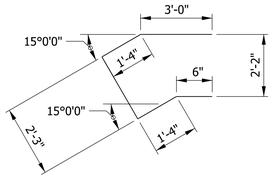
601 x 8'-5"



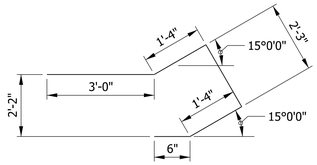
602 x 7'-5"



603 x 3'-8"



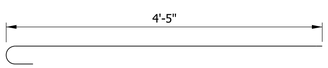
701 x 8'-3"



702 x 8'-5"



#5 x 6'-0"
THREADED TIE BAR ASSEMBLY



451 x 4'-11"

BILL OF MATERIALS
SUPERSTRUCTURE

EPOXY COATED REINFORCING BARS

MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
701	6	8'-3"	
702	6	8'-5"	
#7	12	25'-3"	
#7	48	6'-9"	
TOTAL #7			1486
601	96	8'-5"	
602	110	7'-5"	
603	52	3'-8"	
#6	280	35'-0"	
#6	70	26'-5"	
#6	30	6'-0"	
#6	12	5'-0"	
TOTAL #6			20,583
#5	802	35'-0"	
#5	68	23'-9"	
#5	530	16'-0"	
#5	4	3'-0"	
TOTAL #5			39,819
451	604	4'-11"	
TOTAL #4			1984
TOTAL EPOXY-COATED REINFORCING BARS			63,872

CONCRETE	
CONCRETE, C, SUPERSTRUCTURE	202.1 CYS

MISCELLANEOUS	
THREADED TIE BAR ASSEMBLY, EPOXY COATED, #5	48 EACH
FIELD DRILLED HOLE IN CONCRETE	52 EACH
GEOTEXTILE FOR RIPRAP TYPE 1B	96 SYS
SURFACE SEAL (7659 SFT)	1 LS
AGGREGATE FOR END BENT BACKFILL	26 CYS

NOTES

1. FOR REINFORCING BAR NOTES, SEE STD. DWG. E 703-BRST-01

SKYLER R.P. COOMBS — REGISTERED — No. 11200658 — STATE OF INDIANA — PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MRS 06/2017 | DRAWN: MRS 06/2017

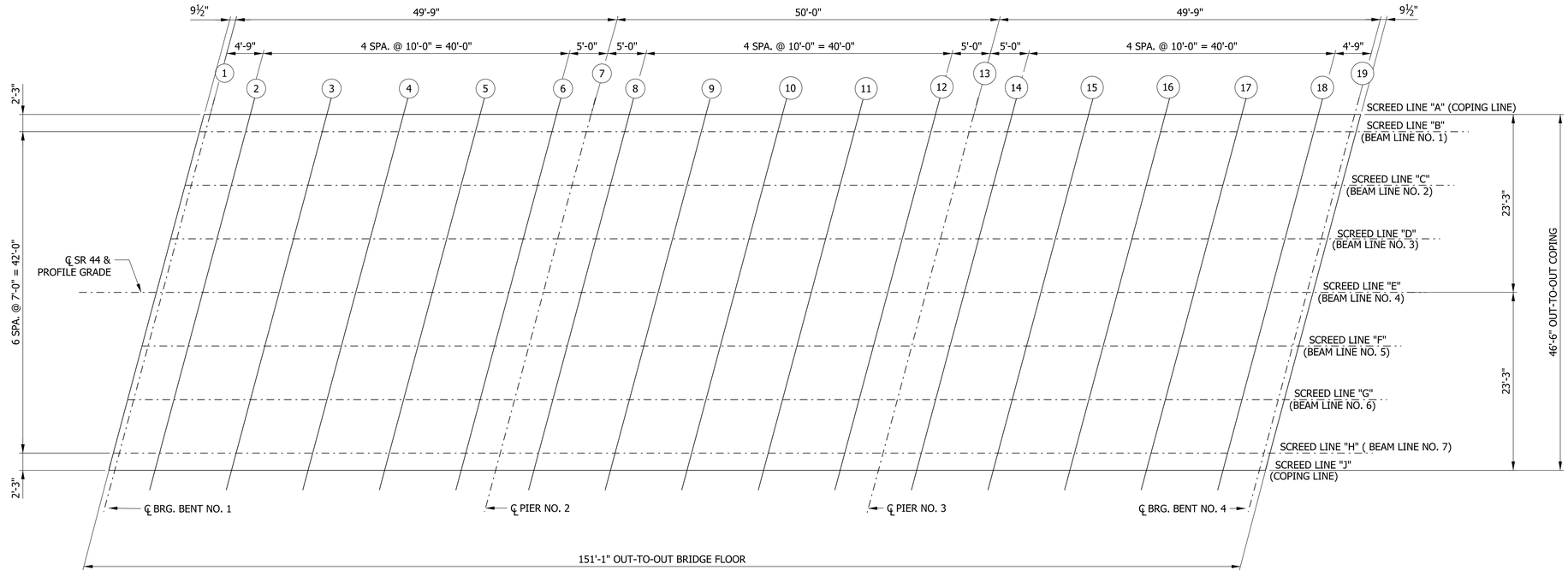
CHECKED: SRPC 06/2017 | CHECKED: SRPC 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

SUPERSTRUCTURE DETAILS

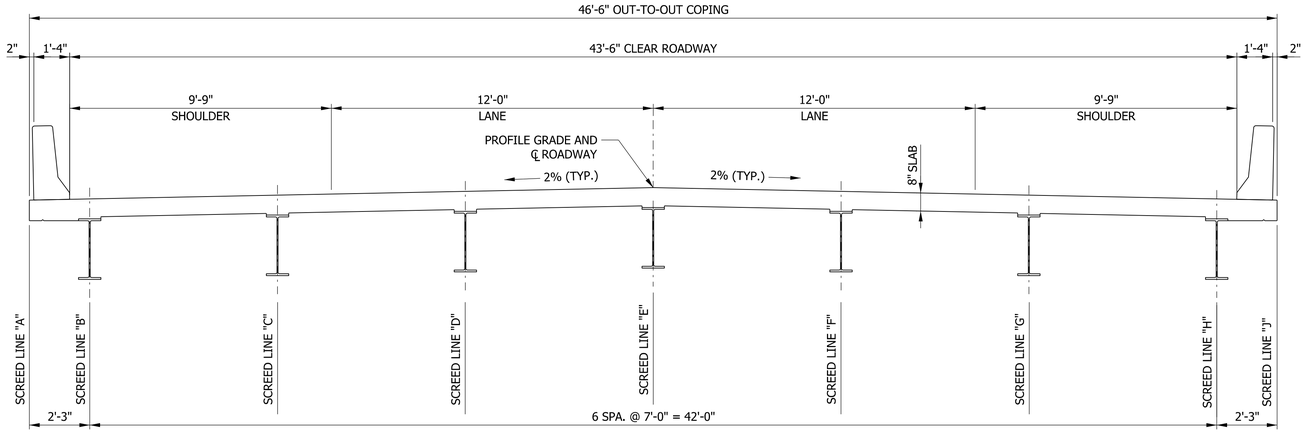
SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	28 of 34
CONTRACT	PROJECT
B-38321	1500342

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model: bill of materials
file: \\indot\1289projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-edb01.dgn



SCREED PLAN

SCALE: 1/8" = 1'-0"



TYPICAL SECTION

SCALE: 3/8" = 1'-0"

NOTES

1. FOR TABLE OF SCREEDS, SEE SHEET 30.

SKYLER R.P. COMBS
REGISTERED
No. 11200658
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MRS 06/2017 | DRAWN: DCK 06/2017

CHECKED: SRPC 06/2017 | CHECKED: MRS 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

SCREEDS

SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	29 of 34
CONTRACT	PROJECT
B-38321	1500342

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6/7/2017 9:42:24 am
model:Screeds
file: \\indw001\289projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-screeds01.dgn

TABLE OF SCREED ELEVATIONS																				
POINT	CONDITION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
A	ELEVATION TOP OF COPING FORM	907.765	907.810	907.895	907.955	907.990	908.005	908.020	908.035	908.070	908.095	908.115	908.125	908.135	908.145	908.170	908.185	908.170	908.135	908.110
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
B	ELEVATION TOP OF SCREED	907.805	907.855	907.935	907.995	908.030	908.050	908.060	908.075	908.110	908.140	908.160	908.170	908.180	908.190	908.215	908.230	908.220	908.180	908.155
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
C	ELEVATION TOP OF SCREED	907.935	907.985	908.075	908.135	908.165	908.185	908.195	908.210	908.245	908.275	908.295	908.305	908.315	908.330	908.360	908.380	908.365	908.325	908.300
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
D	ELEVATION TOP OF SCREED	908.060	908.110	908.205	908.265	908.300	908.315	908.325	908.340	908.380	908.410	908.430	908.445	908.455	908.470	908.500	908.520	908.510	908.470	908.445
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
E	ELEVATION TOP OF SCREED	908.190	908.240	908.330	908.395	908.430	908.450	908.460	908.475	908.515	908.545	908.565	908.580	908.590	908.605	908.640	908.660	908.650	908.610	908.585
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
F	ELEVATION TOP OF SCREED	908.035	908.085	908.180	908.245	908.280	908.300	908.310	908.330	908.370	908.400	908.425	908.440	908.450	908.465	908.500	908.520	908.510	908.475	908.450
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
G	ELEVATION TOP OF SCREED	907.885	907.935	908.030	908.095	908.130	908.150	908.165	908.180	908.220	908.255	908.280	908.295	908.305	908.320	908.360	908.380	908.370	908.335	908.310
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
H	ELEVATION TOP OF SCREED	907.730	907.780	907.870	907.935	907.975	908.000	908.015	908.035	908.075	908.110	908.135	908.150	908.165	908.180	908.210	908.230	908.225	908.195	908.175
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			
J	ELEVATION TOP OF COPING FORM	907.680	907.730	907.820	907.885	907.925	907.955	907.970	907.985	908.030	908.065	908.090	908.105	908.115	908.135	908.165	908.185	908.180	908.150	908.130
	ELEVATION - TOP OF BEAM																			
	DISTANCE - TOP OF BEAM TO TOP OF SCREED																			

NOTES

1. AFTER THE BEAMS ARE SET, TAKE ELEVATIONS AT ALL SCREED POINTS ON TOP OF THE BEAMS. ENTER THESE ELEVATIONS IN THE TABLE. SUBTRACT THESE ELEVATIONS FROM THE TABULATED ELEVATIONS AND USE THE RESULTING DIMENSIONS AS THE HEIGHT FOR SETTING SCREED FORMS ABOVE THESE POINTS. THESE DIMENSIONS REMAIN CONSTANT REGARDLESS OF HOW MUCH OR IN WHAT ORDER THE CONCRETE IS POURED.
2. DO NOT SET SCREED FORMS BY LEVELING.
3. NO CONCRETE IN THE FLOOR SLAB SHALL BE POURED UNTIL THE ABOVE OPERATIONS ARE COMPLETED AND APPROVED BY THE ENGINEER.
4. SCREED ELEVATIONS SHOWN IN THE TABLE INCLUDE AN ALLOWANCE FOR CONCRETE DEAD LOAD DEFLECTIONS.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER 08/22/2017 DATE

DESIGNED: MRS 06/2017 DRAWN: MRS 06/2017

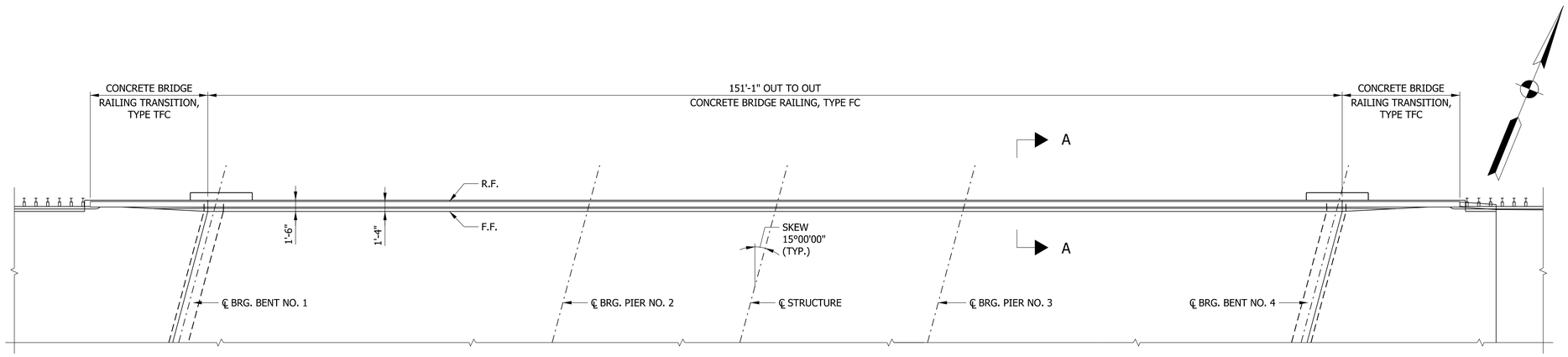
CHECKED: SRPC 06/2017 CHECKED: DCK 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

SCREEDS

SCALE	BRIDGE FILE
N/A	044-70-06634 A
VERTICAL SCALE	DESIGNATION
N/A	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	30 of 34
CONTRACT	PROJECT
B-38321	1500342

mstocker
8/22/2017 3:21:09 pm
model: Screeds table
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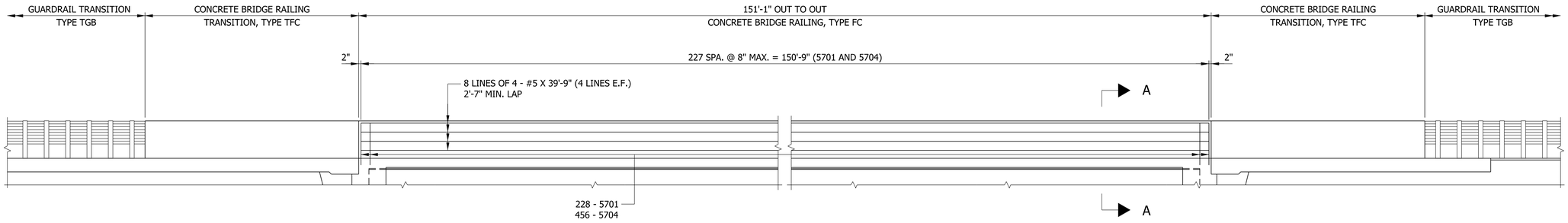
RAILING PLAN

SCALE ⅛" = 1'-0"
(NORTH RAIL SHOWN, SOUTH RAIL SAME)

BILL OF MATERIALS

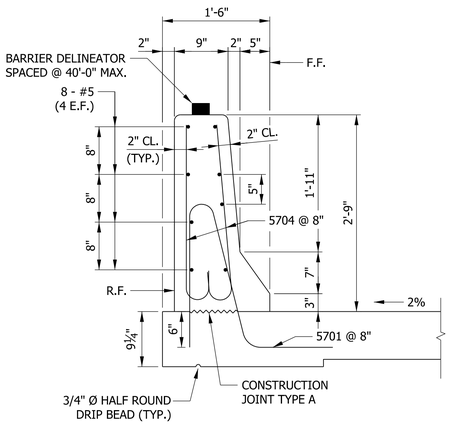
CONCRETE BRIDGE RAILING
(QUANTITES FOR NORTH RAIL, SOUTH RAIL SAME)

EPOXY COATED REINFORCING			
MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
5701	228	5'-6"	
5704	456	3'-0"	
#5	32	39'-9"	
	TOTAL #5		4062
CONCRETE BRIDGE RAILING TRANSITION, TYPE TFC (2 x 551 LBS)			1102
TOTAL EPOXY COATED REINFORCING BARS			5164
CONCRETE			
RAILING, CONCRETE, FC			14.5 CYS
MISCELLANEOUS			
CONCRETE BRIDGE RAILING, TRANSITION, TYPE TFC			2 EACH
SURFACE SEAL (1166 SFT)			1 LS
BARRIER DELINEATOR			6 EACH



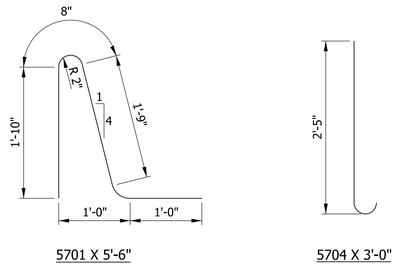
ELEVATION

SCALE ¼" = 1'-0"
(NORTH RAIL SHOWN, SOUTH RAIL SAME)



SECTION A-A

SCALE 1" = 1'-0"



BAR BENDING DETAILS

NOT TO SCALE

LEGEND

R.F. DENOTES REAR FACE

F.F. DENOTES FRONT FACE

E.F. DENOTES EACH FACE

NOTES:

1. ALL REINFORCING BARS SHALL BE EPOXY COATED.
2. FOR ADDITIONAL REINFORCING BAR NOTES AND BAR BENDING DETAILS, SEE STD. DWG. E 703-BRST-01.
3. FOR ADDITIONAL CONCRETE BRIDGE RAILING DETAILS, SEE STD. DWG. E 706-BRSF-01 & -03.
4. FOR BRIDGE RAILING TRANSITION TYPE TFC DETAILS, SEE STD. DWG. E 706-TTFC-01 THROUGH -03.
5. FOR GUARDRAIL TRANSITION TYPE TGB DETAILS, SEE STD. DWG. E 601-TTGB-01 THROUGH -05.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

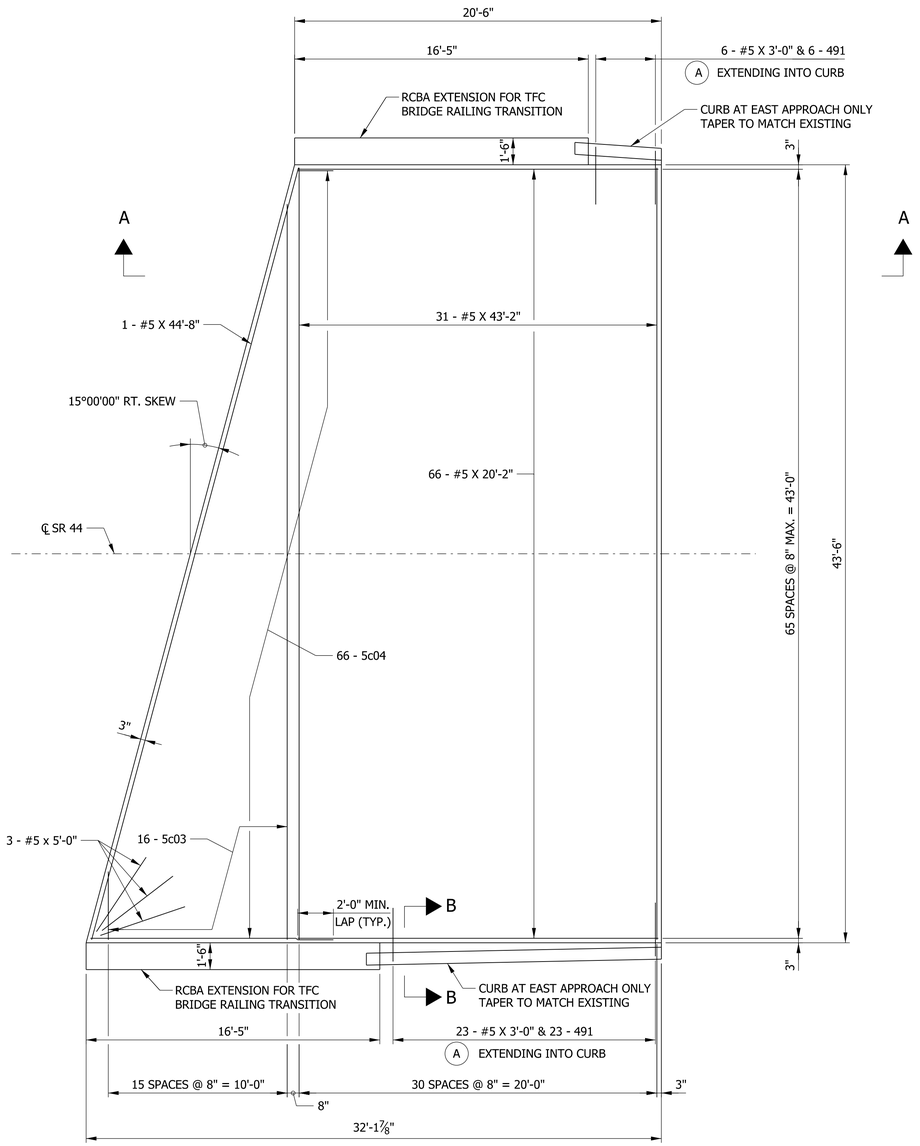
DESIGNED: MEW 05/2017 DRAWN: MEW 05/2017
CHECKED: DCK 05/2017 CHECKED: SRPC 05/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

RAILING DETAILS

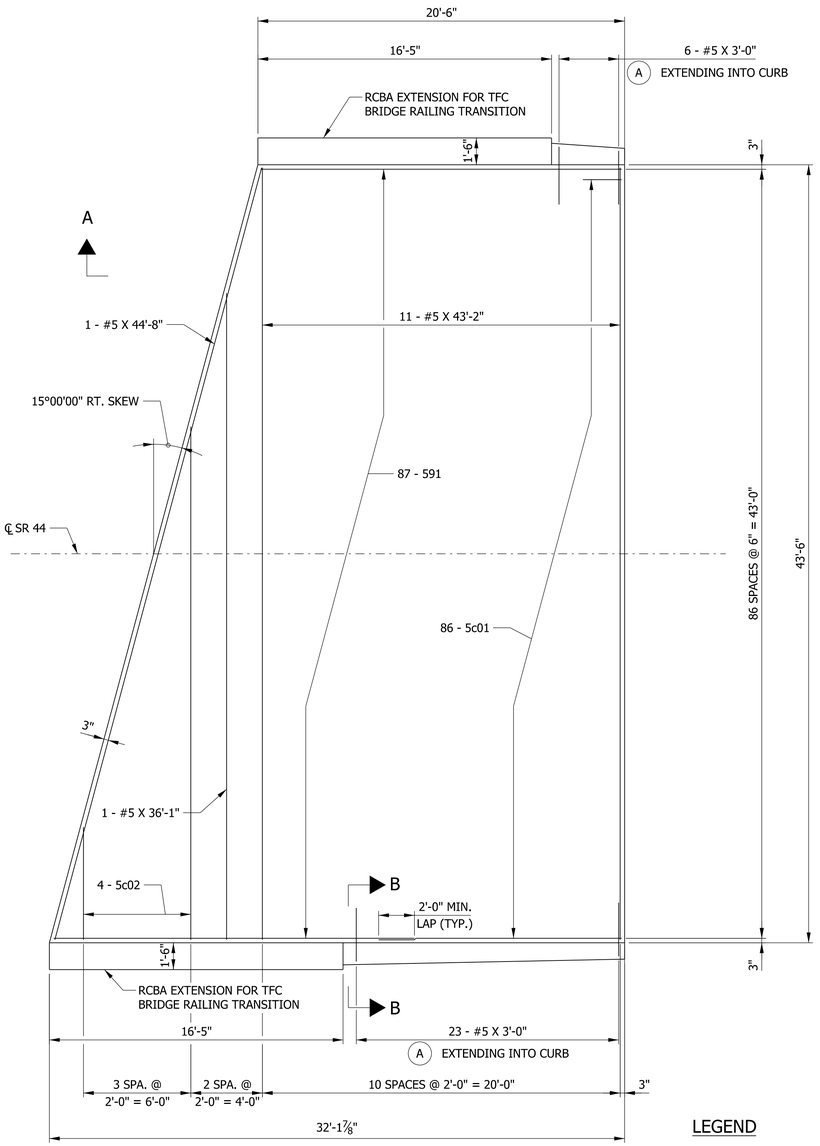
SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	31 of 34
CONTRACT	PROJECT
B-38321	1500342

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model: Railing details
file: \\indw001\289projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-cbd01.dgn



EAST APPROACH SLAB - TOP REINFORCEMENT

(WEST APPROACH SLAB SAME BY 180°)



EAST APPROACH SLAB - BOTTOM REINFORCEMENT

(WEST APPROACH SLAB SAME BY 180°)

LEGEND

(A) BARS ARE PRESENT IN EAST APPROACH SLAB ONLY. SPACE AT 8" MAX. WITH #5 TOP TRANSVERSE REINFORCING. EXTEND INTO CURB.

SKYLER R.P. COOMBS — REGISTERED — No. 11200658 — STATE OF INDIANA — PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MEW 05/2017 DRAWN: MEW 05/2017

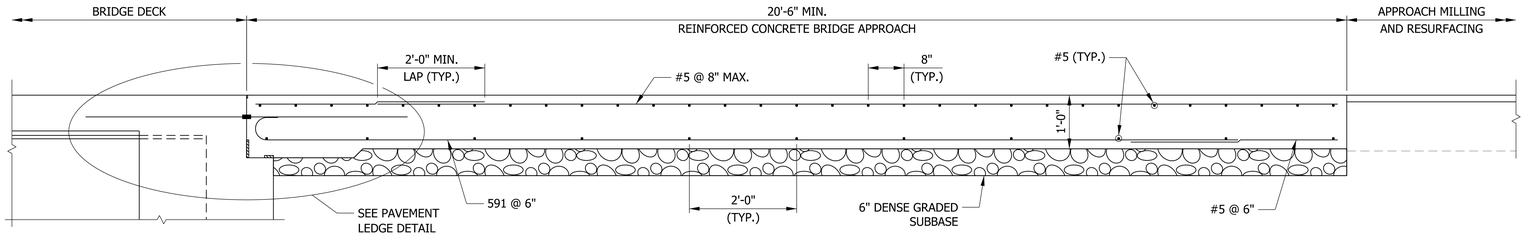
CHECKED: MRS 06/2017 CHECKED: MRS 06/2017

INDIANA DEPARTMENT OF TRANSPORTATION

APPROACH SLAB DETAILS

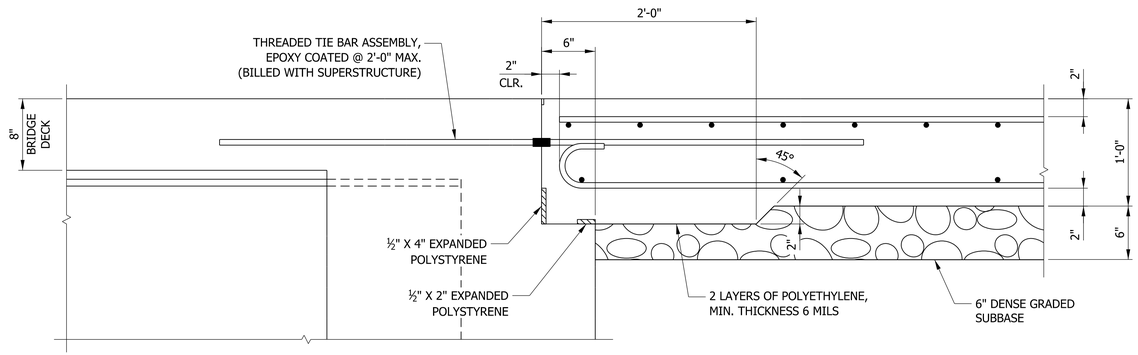
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⅛" = 1'-0"	044-70-06634 A
VERTICAL SCALE	DESIGNATION
¼" = 1'-0"	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	32 of 34
CONTRACT	PROJECT
B-38321	1500342

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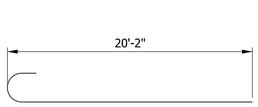
SECTION A-A

SCALE: ¾" = 1'-0"

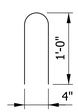


PAVEMENT LEDGE DETAIL

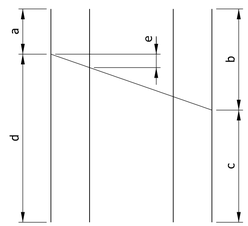
SCALE: 1½" = 1'-0"



591 x 20'-9"



491 x 2'-2"



CUTTING GROUP	DESCRIPTION	a	b	c	d	e
5c01	43 - #5 X 15'-9" MAKES 86 BARS	2'-2"	7'-5"	8'-4"	13'-7"	0'-1 1/2"
5c02	2 - #5 X 34'-11" MAKES 4 BARS	6'-3"	13'-8 1/2"	21'-2 1/2"	28'-8"	7'-5 1/2"
5c03	8 - #5 X 44'-10" MAKES 16 BARS	3'-9"	21'-2"	23'-8"	41'-1"	2'-5 7/8"
5c04	33 - #5 X 15'-7" MAKES 66 BARS	2'-0"	7'-8"	7'-11"	13'-7"	0'-2 1/8"

BILL OF MATERIALS

QUANTITIES FOR WEST APPROACH SLAB

EPOXY COATED REINFORCING BARS

MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
5c01	43	15'-9"	
5c02	2	34'-11"	
5c03	8	44'-10"	
5c04	33	15'-7"	
591	87	20'-9"	
#5	42	43'-2"	
#5	2	44'-8"	
#5	1	36'-1"	
#5	66	20'-2"	
#5	3	5'-0"	
	TOTAL #5		6999
TOTAL FROM RCBA EXTENSION (269 LBS X 2)			538
TOTAL EPOXY COATED REINFORCING BARS			7537

CONCRETE

REINF. CONCRETE BRIDGE APPROACH, 12"	133 SYS

MISCELLANEOUS

DENSE GRADED SUBBASE, 6"	23 CYS
SURFACE SEAL (1200 SFT)	1 LS

BILL OF MATERIALS

QUANTITIES FOR EAST APPROACH SLAB

EPOXY COATED REINFORCING BARS

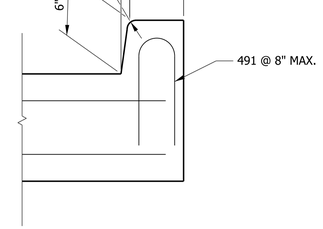
MARK OR SIZE	NO. OF BARS	LENGTH (FT.-IN.)	WEIGHT (LBS.)
5c01	43	15'-9"	
5c02	2	34'-11"	
5c03	8	44'-10"	
5c04	33	15'-7"	
591	87	20'-9"	
#5	42	43'-2"	
#5	2	44'-8"	
#5	1	36'-1"	
#5	66	20'-2"	
#5	3	5'-0"	
#5	58	3'-0"	
	TOTAL #5		7180
491	29	2'-2"	
	TOTAL #4		42
TOTAL FROM RCBA EXTENSION (269 LBS X 2)			538
TOTAL EPOXY COATED REINFORCING BARS			7760

CONCRETE

REINF. CONCRETE BRIDGE APPROACH, 12"	136 SYS

MISCELLANEOUS

DENSE GRADED SUBBASE, 6"	23 CYS
SURFACE SEAL (1220 SFT)	1 LS



SECTION B-B

SCALE: 1 ½" = 1'-0"

NOTES:

1. FOR ADDITIONAL REINFORCING BAR NOTES AND DETAILS, SEE STANDARD DRAWING E703-BRST-01.
2. FOR REINFORCING DETAILS OF RCBA EXTENSION FOR TFC, SEE STANDARD DRAWING E609-TBAE-01. THICKNESS SHALL MATCH APPROACH SLAB THICKNESS OF 12".
3. ALL REINFORCING TO BE EPOXY COATED.
4. FOR DETAILS OF TYPE I-A JOINT, SEE STANDARD DRAWING E609-BRJT-01.

SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: MEW 05/2017 DRAWN: MEW 05/2017

CHECKED: MRS 06/2017 CHECKED: MRS 06/2017

INDIANA
DEPARTMENT OF TRANSPORTATION

APPROACH SLAB DETAILS

SCALE	BRIDGE FILE
AS SHOWN	044-70-06634 A
VERTICAL SCALE	DESIGNATION
AS SHOWN	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	33 of 34
CONTRACT	PROJECT
B-38321	1500342

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file: \\indw001\299projects\64359 indot gf bridge rehab\cadd\b13 sr 44 over mud creek\cds\64359-s-br-asd01.dgn

SUMMARY OF BRIDGE QUANTITIES

ITEM	INSPECTION HOLE 3FT DEEP OR LESS	PAVEMENT REMOVAL	CURB, CONCRETE, REMOVE	EXCAVATION, COMMON	PUMP AROUND	EXCAVATION, FOUNDATION, UNCLASSIFIED	AGGREGATE FOR END BENT BACKFILL	DENSE GRADED SUBBASE	BARRIER DELINEATOR	REINFORCED CONCRETE BRIDGE APPROACH, 12"	RIPRAP, REVETMENT	GEOTEXTILES FOR RIPRAP TYPE 1A	GEOTEXTILES FOR RIPRAP TYPE 1A	SODDING	FIELD DRILLED HOLE IN CONCRETE	REINF. BARS, EPOXY COATED	CONCRETE, C, SUB-STRUCTURE	THREADED TIE BAR ASSEMBLY EPOXY COATED	CONCRETE, C, SUPER-STRUCTURE	RAILING, CONCRETE FC	CONCRETE BRIDGE RAILING TRANSITION, TFC	GROUT, NON-SHRINK	SURFACE SEAL	PATCHING CONCRETE STRUCTURES	STRUCTURAL STEEL
	EACH	SYS	LFT	CYS	EACH	CYS	CYS	CYS	EACH	SYS	TON	SYS	SYS	SYS	EACH	LBS	CYS	EACH	CYS	CYS	EACH	CYS	SFT *	SFT	LBS *
STRUCTURE NO. 044-70-06634 A	2	128	20	70	1	40	26	46	12	269	50	93	96	62	188	90,487	3.8	48	202.1	29.0	4	0.2	12800	70	117,950
TOTAL	2	128	20	70	1	40	26	46	12	269	50	93	96	62	188	90,487	3.8	48	202.1	29.0	4	0.2	12800	70	117,950

* For information only. Estimated quantity paid for as lump sum per bridge.

SUMMARY OF BRIDGE QUANTITIES

ITEM	CASTING, ADJUST TO GRADE	LINE, PAINT, SOLID, WHITE, 4"	LINE, PAINT, SOLID, YELLOW, 4"	TRANSVERSE MARKING, PAINT, STOP LINE, WHITE, 24"	SNOWPLOWABLE RAISED PVMT MARKER, REMOVE	SNOWPLOWABLE RAISED PVMT MARKER
	EACH	LFT	LFT	LFT	EACH	EACH
STRUCTURE NO. 044-70-06634 A	2	907	842	12	9	14
TOTAL	2	907	842	12	9	14

PAVEMENT QUANTITIES AND APPROACH TABLE

LOCATION	DESCRIPTION (APPROACH TYPE OR CLASS)	HMA FOR ROADS		LIQUID ASPHALT SEALANT	JOINT ADHESIVE, SURFACE	JOINT ADHESIVE, INTERMEDIATE	MILLING TRANSITION	ASPHALT MATERIAL	
		QC/QA-HMA, 3, 70, SURFACE, 9.5mm	QC/QA-HMA, 3, 70, INTERMEDIATE, 19.0mm					PRIME COAT	TACK COAT
		TON	TON	LFT	LFT	LFT	SYS	SYS	SYS
SR 44 LINE "A"		123	205	937	937	937	1479		2958
	TOTALS	123	205	937	937	937	1479		2958

GUARDRAIL SUMMARY TABLE

LOCATION		LEFT	RIGHT	GUARDRAIL, TRANSITION, TYPE TGB	GUARDRAIL, CONNECTOR SYSTEM, W-BEAM CURVED, 2, MODIFIED	GUARDRAIL, TERMINAL SYSTEM, W-BEAM CURVED, 2	GUARDRAIL, END TREATMENT, OS	GUARDRAIL W-BEAM, 6 FT 3IN SPACING	GUARDRAIL REMOVE
FROM STATION	TO STATION								
				EACH	EACH	EACH	EACH	LFT	LFT
616+08.46	616+98.56		X	1	1		1	12.5	90
615+97.70	617+10.22	X		1			1	37.5	100
618+79.78	619+67.24		X	1			1	12.5	55
618+91.43	619+50.29	X		1		1		25.0	85
	TOTALS			4	1	1	3	88	330

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SKYLER R.P. COMBS REGISTERED No. 11200658 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL — DESIGN ENGINEER — 08/22/2017 DATE

DESIGNED: SRPC 08/2016 | DRAWN: SRPC 08/2016

CHECKED: CDC 08/2016 | CHECKED: CDC 08/2016

INDIANA
DEPARTMENT OF TRANSPORTATION

BRIDGE SUMMARY

SCALE	BRIDGE FILE
N/A	044-70-06634 A
VERTICAL SCALE	DESIGNATION
N/A	1500342
SURVEY BOOK	SHEETS
ELECTRONIC	34 of 34
CONTRACT	PROJECT
B-38321	1500342